

Monetary Policy and the Predictability of Nominal Exchange Rates

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This article studies how the monetary policy regime affects the relative importance of nominal exchange rates and inflation rates in shaping the response of real exchange rates to shocks. We document two facts about inflation-targeting countries. First, the current real exchange rate predicts future changes in the nominal exchange rate. Second, the real exchange rate is a poor predictor of future inflation rates. We estimate a medium-size, open-economy DSGE model that accounts quantitatively for these facts as well as other empirical properties of real and nominal exchange rates. The key estimated shocks that drive the dynamics of exchange rates and their covariance with inflation are disturbances to the foreign demand for dollar-denominated bonds.

Key words: Monetary policy, Nominal exchange rate predictability

JEL Codes: E52, F31, F41

1. INTRODUCTION

This article studies how monetary policy affects the relative importance of nominal exchange rates (*NERs*) and inflation rates in shaping the response of real exchange rates (*RERs*) to shocks. Our analysis has four components. First, we document key empirical regularities that govern the relation between the current *RER* and future values of inflation and the *NER*. We then show that these regularities depend critically on the monetary policy regime in effect. Second, we provide a simple theory that explains these regularities. Third, we identify the key shocks that have driven the U.S. *RER*, *NER*, and their relation to inflation under an inflation-targeting regime. We do so using an estimated three-country DSGE model of the world economy. Finally, we show how

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the adjustment to these shocks would have occurred under alternative monetary policy regimes. Building on a long empirical literature discussed below, we document two facts about *RERs* and *NERs* for a set of inflation-targeting, benchmark countries. To describe our results, it is useful to define the *RER* as the price of the foreign-consumption basket in units of the home-consumption basket and the *NER* as the price of the foreign currency in units of the home currency. Our first fact is that the current *RER* is highly negatively correlated with future changes in the *NER* at horizons greater than two years. This correlation is stronger the longer the horizon. Our second fact is that the *RER* is virtually uncorrelated with future inflation rates at all horizons. Taken together, these facts imply that the *RER* adjusts to shocks in the medium and long run overwhelmingly through changes in the *NER*, not through differential inflation rates.

Critically, these facts depend on the monetary policy regime in effect. To show this dependency, we re-do our analysis for China which is on a quasi-fixed exchange rate regime versus the U.S. dollar; for Hong Kong, which has a fixed exchange rate, versus the U.S. dollar; and for the euro-area countries, which have fixed exchange rates with each other. In all of these cases, the current *RER* is highly negatively correlated with future relative inflation rates. In contrast to our benchmark countries, the *RER* adjusts overwhelmingly through predictable inflation differentials.

We also redo our analysis for a set of countries that adopted inflation targeting after our benchmark countries (around the year 2000). This set of countries consists of Brazil, Chile, Colombia, Indonesia, Israel, Mexico, South Korea, and Thailand. We show that when these countries adopted inflation targeting, the dynamic co-movements of the *NER*, the *RER*, and inflation became qualitatively and quantitatively similar to those in our benchmark countries. This type of sensitivity to the monetary policy regime is precisely what we would expect given the Lucas (1976) critique.

A natural question is whether our findings are spurious in the sense that they might primarily reflect small sample sizes and persistent *RERs*.¹ To address these concerns, we show that *out-of-sample* forecasts of the *NER* based on the *RER* beat a random-walk forecast at medium and long horizons. We argue that this finding is extremely unlikely if the *NER* is not predictable, regardless of whether the *RER* is stationary. This result strongly supports the view that our empirical findings are not spurious.

Having established our key facts, we turn to the underlying economics. We begin with a simple two-country endowment model to explain the core mechanisms that drive our empirical results. These mechanisms depend on two key assumptions: the monetary authority follows an inflation-targeting policy like a Taylor rule and the *RER* changes in response to various shocks. In principle, this response can occur for many reasons, such as introducing home bias, non-tradable goods, or sticky prices. In our simple model, changes in the *RER* in response to endowment shocks result from home bias in consumption.

The intuition for the mechanisms that generate the relevant empirical regularities in our simple model is as follows. Consider a persistent fall in the domestic endowment. This shock leads to a rise in the price of the domestic good. Consumers combine a domestic and foreign good into a final consumption good. Because of home bias, the domestic good has a higher weight than the foreign good in the domestic consumption basket. So, after the shock, the price of the foreign consumption basket in units of the home consumption basket falls—i.e. the *RER* falls. Because the home and foreign monetary authorities follow a Taylor rule that keeps inflation relatively stable, the *RER* must adjust through movements in the *NER*. This result holds, even though prices are perfectly flexible in our simple model.

1. Similar concerns lie at the heart of ongoing debates about the predictability of the equity premium based on variables like the price–dividend ratio (see Stambaugh, 1999; Boudoukh et al., 2006; Cochrane, 2007).

We next turn to the question, what shocks and frictions account *quantitatively* for the movements in the *RER* and the *NER* as well as their covariance with inflation? To answer this question, we construct and estimate a medium-scale, three-country (U.S., Germany, and the rest of the world) DSGE model. Consistent with our simple model, we assume that all three countries follow a Taylor rule and have home bias in consumption. Production requires labour and capital services. Nominal prices and wages are set subject to Calvo-style frictions. There is local currency pricing, and asset markets are incomplete: the only internationally traded asset is a dollar-denominated bond.² We allow for many different types of shocks to affect economic agents and economic activity. In this way, we can estimate which shocks are important in practice.

According to our estimated model, the key shock that drives the correlation between the current *RER* and future changes in the *NER* are shocks to foreign demand for dollar-denominated bonds. In particular, roughly 75% of the relevant covariances are driven by this shock. Significantly, this shock is *not* an important driver of either U.S. or German output fluctuations in our model. These findings are firmly in the tradition of papers like [Kollmann \(2005\)](#), [Gabaix and Maggiori \(2015\)](#), and [Itskhoki and Mukhin \(2017\)](#), who stress disturbances emanating from financial markets—*e.g.* liquidity shocks—as drivers of exchange rate volatility.

Our results are interesting to the extent that our model is a credible representation of the data. Our estimated model has a number of quantitative properties that lend support to its credibility. First, it accounts quantitatively for the correlations between the current *RER*, future inflation rates and changes in the *NER*. Second, it accounts quantitatively for the volatility of the *RER* and changes in the *NER*, the persistence of the *RER*, and the high correlation between the *RER* and the *NER*. Third, it accounts quantitatively for the failure of uncovered interest rate parity (UIP) as measured by the estimated slope coefficient in a regression of the change in the *NER* on the interest-rate differential (see [Fama, 1984](#)). The model is also consistent with [Backus and Smith \(1993\)](#)'s finding of a low correlation between relative consumption and the *RER*, a fact that is inconsistent with a wide class of models.

We use our estimated model to ask the counterfactual question, given the estimated distribution of the shocks that occurred in our sample period, how would the economy have reacted under two alternative monetary policy regimes: a nominal exchange rate targeting regime and a regime with capital controls? Under a nominal exchange rate targeting regime, the *NER* plays a substantially smaller role and differential inflation a larger role in re-establishing long-run purchasing power parity (PPP). Additionally, the *RER* is more persistent under nominal exchange rate targeting than under inflation targeting. By contrast, the capital controls we consider have a small effect on the relevant equilibrium properties of the economy.

Our article is organized as follows. Section 2 briefly discusses how our article relates to the literature. Section 3 contains our empirical results. Section 4 offers intuition for the drivers of our empirical results using a simple model. In Section 5, we present a medium-scale DSGE model that we estimate using full-information Bayesian methods. In Section 6, we show that the estimated medium-scale DSGE model can explain our correlation estimates reported in Section 3. We discuss which estimated shocks drive those correlations and the movements in the *RER*. In addition, we explore how the *RER* would have adjusted to shocks under alternative monetary policy regimes. Section 7 concludes.

2. [Maggiori et al. \(2020\)](#) document the fact that almost all internationally traded bonds are denominated in U.S. dollars.

2. RELATION TO THE LITERATURE

Our work is related to four strands of the literature. The first strand studies the medium- and long-run predictability of *NERs*. For example, [Mark \(1995\)](#) and [Engel *et al.* \(2007\)](#) find evidence of predictability at medium and long horizons. See [Rossi \(2013\)](#) for a recent survey. A closely associated body of work demonstrates that relative PPP holds in the long run so that *RERs* are mean reverting and therefore predictable (see [Rogoff, 1996](#) and [Taylor and Taylor, 2004](#) for a review). In general, the predictability of the *RER*, or the fact that relative PPP holds in the long run, does not imply that the *NER* is predictable. For example, if monetary policy seeks to limit the volatility of the *NER*, the *RER* converges to its unconditional mean primarily via inflation differentials rather than via sustained, predictable movements in the *NER*.

Our contribution to this literature is to draw a tight connection between the usefulness of the current *RER* in predicting the future *NER* (in and out of sample) and the monetary policy regime in effect. We also show that our predictability results are robust to the possibility that the *RER* is not stationary. Finally, we construct and estimate a medium-scale, open-economy DSGE model that is quantitatively consistent with the dynamic correlations that drive the predictability of the *NER* in inflation-targeting regimes.

Our empirical predictability results pertain to medium and long horizons and are distinct from the voluminous literature started by [Meese and Rogoff \(1983\)](#) on the out-of-sample predictability of the *NER* at short horizons (up to one year). Our predictability results are also distinct from [Mussa \(1986\)](#)'s famous demonstration that contemporaneous changes in the *NER* and the *RER* are highly correlated.

The second relevant strand of the literature discusses the importance of the monetary regime for the behaviour of the *RER*. See, for example, [Mussa \(1986\)](#), [Baxter and Stockman \(1989\)](#), [Henderson and McKibbin \(1993\)](#), [Sarno and Valente \(2006\)](#), [Engel *et al.* \(2007\)](#), and [Engel \(2019\)](#). Our contribution is to focus on the behaviour of the *RER* under an inflation-targeting regime per se and to use our estimated DSGE model to study how the *NER*, the *RER*, and the real economy would have evolved under alternative monetary regimes.

The third relevant strand of the literature studies a myriad of failures of standard open-economy macro models, including the failure of UIP and the disconnect between the *RER* and relative consumption across countries. To explain a sub-set of these failures, [McCallum \(1994\)](#), [Kollmann \(2005\)](#), [Christiano *et al.* \(2011\)](#), and [Engel and Wu \(2019\)](#) stress the importance of shocks to the relationship defining the UIP condition. The latter three papers argue on quantitative grounds that UIP shocks are very important sources of *NER* and *RER* fluctuations. There are various differences between our estimated model and the ones considered in those papers. But the critical differences are that those papers do not allow for shocks to the foreign demand for dollar-denominated bonds and do not analyse the predictability of nominal exchange rates.

In contemporaneous work, [Itskhoki and Mukhin \(2017\)](#) argue that financial shocks, like shocks to the global demand for dollar-denominated bonds, can explain the failure of UIP, the low correlation between the *RER* and relative consumption across countries, and the more general disconnect between exchange rates and many real variables. [Jiang *et al.* \(2019\)](#) also stress the importance of these types of shocks as drivers of U.S. *NERs*. Our contribution relative to these papers is 2-fold. First, we document empirically the predictability of the *NER* under inflation-targeting regimes. Second, in a medium-scale, open-economy DSGE model, we formally estimate which shocks are the key drivers of the *RER* and the *NER* under an inflation-targeting regime. Because we formally estimate our model, we can use it to investigate how exchange rates and inflation would have behaved under alternative policy regimes. We consider two such regimes: *NER* targeting and a regime that imposes a form of capital controls.

The fourth strand of related literature pertains to estimated open-economy DSGE models. [Lubik and Schorfheide \(2005\)](#) use Bayesian methods to estimate a small-scale, two-country model for the U.S. and the euro area. These authors focus on the challenges to estimation, including model specification and identification problems. [Kollmann *et al.* \(2016\)](#) estimate a large-scale, three-region model for the U.S., the euro area, and the rest of the world. Those authors focus on the post-crisis divergence in real output growth between the U.S. and the euro area. We estimate a medium-scale, three-region DSGE model and focus on identifying which shocks are the important drivers of *RERs* and *NERs*.

3. SOME EMPIRICAL PROPERTIES OF EXCHANGE RATES

In this section, we present our empirical results regarding *NERs*, *RERs*, and relative inflation rates. We use consumer price indexes for all items and average quarterly *NERs* versus the U.S. dollar.³

3.1. Data

We initially focus on a benchmark group of advanced economies—Australia, Canada, Germany, New Zealand, Sweden, and the U.K. With the exception of Germany, these countries formally adopted inflation targeting before 1997. Our source for adoption dates is [Ilzetzi *et al.* \(2017, 2019\)](#). We limit the set of countries to those that adopted inflation targeting before 1997 because we need to have sufficient data to perform our statistical analyses.

[Bernanke and Mihov \(1997\)](#) argue that, even though Germany never explicitly adopted an inflation-targeting regime, the Bundesbank was in fact targeting inflation. For this reason, we include Germany in our benchmark group.⁴ For Germany, we start the sample period in 1982:Q4, which is the beginning of the period for which, according to [Clarida *et al.* \(1998\)](#), U.S. monetary policy is best characterized as a stable Taylor rule.⁵

We exclude data from 2008:Q4 to the present from our benchmark sample period because short-term U.S. nominal interest rates were at or near their effective lower bound.⁶ [Supplementary Appendix A](#) reports results obtained from starting the sample period in 1973 and extending the sample period after 2008.

We compare results for the benchmark inflation targeters with those for China (from 1994 through 2008), which has been on a quasi-fixed exchange rate vis-à-vis the U.S. dollar, and for Hong Kong (from 1982 through 2008), which has a fixed exchange rate vis-à-vis the U.S. dollar. We also analyse data starting in 1999 for France, Ireland, Italy, Portugal, and Spain where the *RER* and relative inflation rates are defined relative to Germany.

In addition, we report results for a subset of our statistical analyses for countries that became inflation targeters between 1997 and 2002. This set of countries consists of Brazil,

3. Our data for bilateral exchange rates between the U.S. and other countries and consumer price indexes are from the International Monetary Fund's International Financial Statistics database.

4. As in [Engel *et al.* \(2007\)](#), we merge exchange rate data for the German mark and the euro after 1999.

5. The beginning of the sample period for Australia, Canada, Germany, New Zealand, Sweden, and the U.K. is 1993:Q3, 1991:Q2, 1982:Q4, 1990:Q1, 1996:Q1, and 1992:Q4, respectively. Of course, actual monetary policy may in practice deviate from the parsimonious Taylor rule that we consider. For example, policymakers in our benchmark countries may have occasionally intervened in currency markets, but these interventions were not a defining characteristic of their monetary policy regime.

6. See [Amador *et al.* \(2019\)](#) for a discussion of the effect of the zero lower bound on exchange-rate policies.

Chile, Colombia, Israel, Mexico, Norway, Peru, the Philippines, South Africa, South Korea, and Thailand.⁷ We refer to this set of countries as the recent inflation targeters.

3.2. Results for inflation-targeting countries

We define the *RER* for country *i* relative to the U.S. as

$$RER_{i,t} = \frac{NER_{i,t}P_{i,t}}{P_t}, \quad (3.1)$$

where $NER_{i,t}$ is the price of the foreign currency (U.S. dollars per foreign currency unit). The variables P_t and $P_{i,t}$ denote the consumer price index in the U.S. and in country *i*, respectively. We assume that the *RER* is stationary and offer supporting evidence later in this section.⁸ Given this assumption, the *RER* must adjust back to its mean after a shock via changes in the *NER* or changes in relative prices.

Figure 1 displays scatter plots of the $\log(RER_{i,t})$ against $\log(NER_{i,t+h}/NER_{i,t})$ at different horizons, *h*, for our benchmark countries. Two properties of this figure are worth noting. First, consistent with the notion that exchange rates behave like random walks at high frequencies, there is no obvious relationship between the $\log(RER_{i,t})$ and $\log(NER_{i,t+h}/NER_{i,t})$ at a one-year horizon. However, as the horizon expands, the correlation between $\log(RER_{i,t})$ and $\log(NER_{i,t+h}/NER_{i,t})$ rises. The negative relation is very pronounced at the five-year horizon.

3.2.1. Nominal exchange rate regressions. We now discuss results for the benchmark countries based on the following *NER* regression

$$\log\left(\frac{NER_{i,t+h}}{NER_{i,t}}\right) = \alpha_{i,h}^{NER} + \beta_{i,h}^{NER} \log(RER_{i,t}) + \varepsilon_{i,t,t+h}^{NER} \quad (3.2)$$

for country *i* at horizon $h = 1, 2, \dots, H$ years. Panel (a) of Table 1 reports estimates of $\beta_{i,h}^{NER}$, along with standard errors, for the benchmark inflation-targeting countries.⁹ A number of features are worth noting. First, the estimated values of $\beta_{i,h}^{NER}$ are negative for all *h*. Second, the estimated values of $\beta_{i,h}^{NER}$ for all *i* are statistically significant at three-year horizons or longer. Third, in general, the estimated value of $\beta_{i,h}^{NER}$ increases in absolute value for roughly the first five years and then stabilizes.

Taken together, the results in Table 1 strongly support the conclusion that, for our benchmark countries, the current *RER* is highly negatively correlated with changes in future *NER*s at horizons of three or more years. These results are consistent with those obtained by Cheung *et al.* (2004) using a vector error-correction model. One substantive difference between our results and theirs is that, for horizons greater than one year, our point estimates of $\beta_{i,h}^{NER}$ are greater than one, indicating that the *NER* adjusts more than the *RER* over time. This finding reflects the fact that relative inflation rates initially move in the wrong direction for re-establishing long-run relative PPP.

7. The beginning of the inflation-targeting sample period for Brazil, Chile, Colombia, Israel, Mexico, Norway, Peru, the Philippines, South Africa, South Korea, and Thailand is 1999:Q4, 1999:Q3, 1999:Q4, 1997:Q3, 2002:Q1, 2001:Q2, 2002:Q1, 2002:Q1, 2000:Q2, 1998:Q2, and 2000:Q3. We exclude countries that adopted inflation targeting after 2002 or have pre-inflation-targeting data available only for short samples.

8. See Rogoff (1996) for an early discussion of the stationarity of the *RER*.

9. We compute standard errors using an estimator of Newey and West (1987) with the number of lags equal to the forecasting horizon plus eight quarters. If not feasible, we use the sample size minus two quarters.

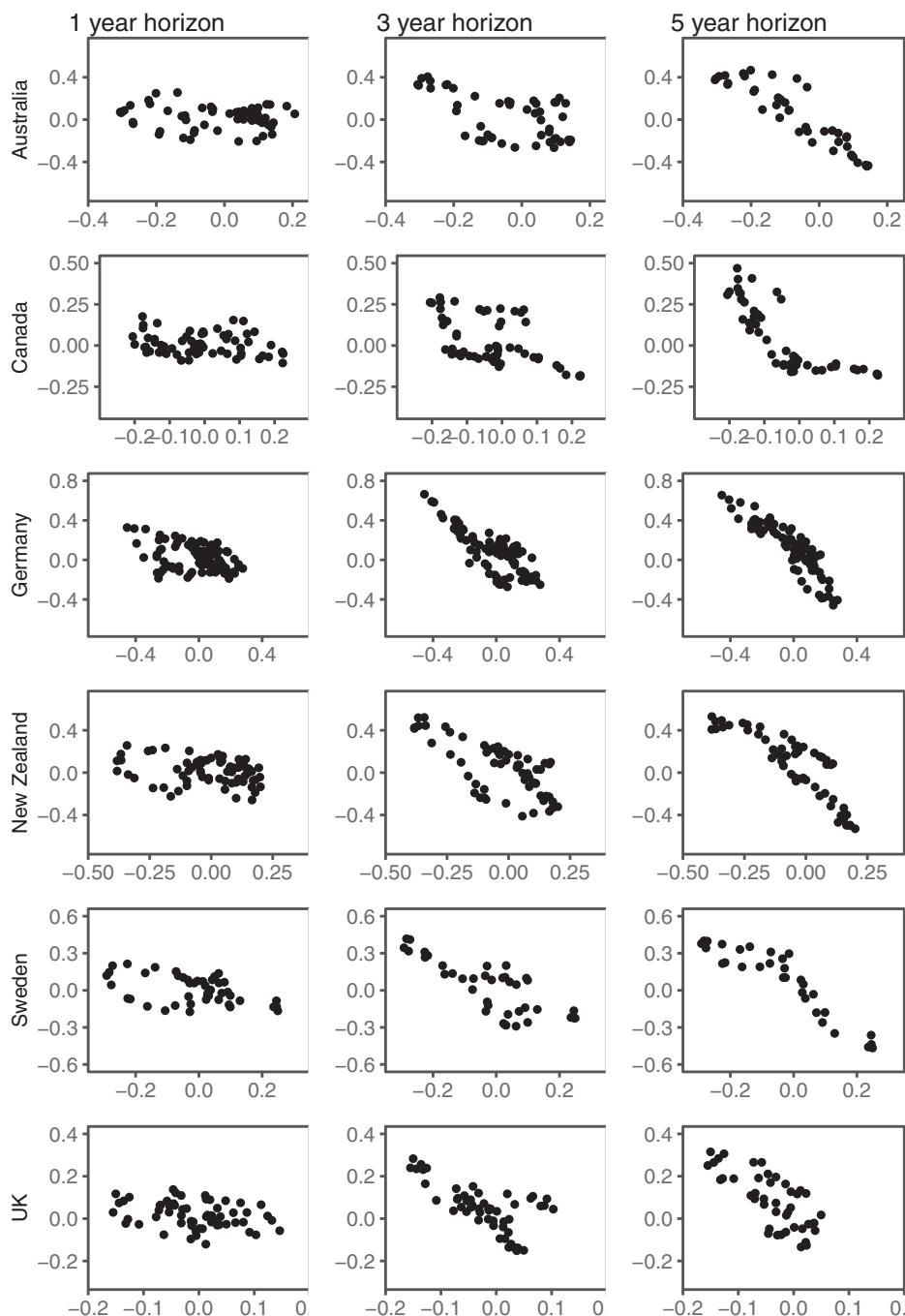


FIGURE 1

NER and RER data. The horizontal axis is the log of the *RER*. The vertical axis is the future change in the log of the *NER* at the specified horizon. *Sources*: Data from International Monetary Fund, International Financial Statistics; authors' calculations.

TABLE 1
Regression results.

	$\beta_{i,h}^{NER}$ Horizon (in years)				$\beta_{i,h}^{\pi}$ Horizon (in years)			
	1	3	5	7	1	3	5	7
(a) Benchmark countries								
Australia	-0.10 (0.17)	-0.84 (0.20)	-1.95 (0.08)	-1.99 (0.14)	-0.04 (0.01)	-0.10 (0.04)	-0.06 (0.04)	-0.03 (0.02)
Canada	-0.11 (0.14)	-0.68 (0.17)	-1.35 (0.30)	-1.74 (0.30)	-0.05 (0.01)	-0.13 (0.01)	-0.14 (0.03)	-0.15 (0.04)
Germany	-0.27 (0.12)	-0.97 (0.10)	-1.34 (0.16)	-1.58 (0.13)	0.02 (0.01)	0.08 (0.03)	0.14 (0.06)	0.14 (0.06)
New Zealand	-0.21 (0.13)	-1.06 (0.13)	-1.62 (0.25)	-1.44 (0.15)	-0.01 (0.01)	-0.06 (0.02)	-0.09 (0.01)	-0.08 (0.02)
Sweden	-0.36 (0.13)	-1.12 (0.13)	-1.57 (0.15)	-1.31 (0.04)	-0.05 (0.02)	-0.06 (0.02)	0.00 (0.02)	-0.00 (0.02)
U.K.	-0.26 (0.10)	-1.09 (0.45)	-1.68 (0.15)	-1.08 (0.44)	-0.02 (0.01)	-0.03 (0.04)	-0.01 (0.05)	-0.08 (0.05)
(b) Managed exchange rates								
China	-0.10 (0.04)	-0.32 (0.13)	-0.40 (0.21)	-0.30 (0.19)	-0.41 (0.17)	-0.90 (0.18)	-1.04 (0.07)	-0.98 (0.01)
Hong Kong	0.04 (0.04)	0.04 (0.04)	0.03 (0.04)	0.02 (0.04)	-0.09 (0.06)	-0.38 (0.14)	-0.79 (0.16)	-1.18 (0.15)
(c) Euro area vis-a-vis Germany								
France					-0.11 (0.09)	-0.77 (0.27)	-1.40 (0.25)	-1.39 (0.22)
Italy					-0.19 (0.05)	-0.54 (0.09)	-0.77 (0.11)	-0.82 (0.13)
Ireland					-0.27 (0.09)	-0.80 (0.05)	-1.12 (0.05)	-1.45 (0.04)
Portugal					-0.24 (0.05)	-0.68 (0.06)	-0.89 (0.05)	-1.05 (0.04)
Spain					-0.16 (0.03)	-0.47 (0.08)	-0.73 (0.08)	-0.93 (0.06)

Sources: Authors' calculations; data from International Monetary Fund, International Financial Statistics.

3.2.2. Relative-price regressions. We now consider results based on the following relative-price regression:

$$\log\left(\frac{P_{i,t+h}/P_{t+h}}{P_{i,t}/P_t}\right) = \alpha_{i,h}^{\pi} + \beta_{i,h}^{\pi} \log(RER_{i,t}) + \varepsilon_{i,t,t+h}^{\pi}. \quad (3.3)$$

This regression quantifies how much of the adjustment in the *RER* occurs via changes in relative rates of inflation across countries. Panel (a) of Table 1 reports our estimates and standard errors for the slope coefficients $\beta_{i,h}^{\pi}$. The key result is that the coefficients are small relative to the estimated values of $\beta_{i,h}^{NER}$. Moreover, the estimated values of β_h^{π} and $\beta_{i,h}^{\pi}$ are not generally statistically different from zero. Taken together, these results suggest that movements in relative prices account for a small fraction of movements in *RERs*.

3.3. Sensitivity to monetary policy

Our basic hypothesis is that the process by which the *RER* adjusts to shocks depends critically on the monetary policy regime. We provide two types of evidence in favour of this hypothesis. First,

we redo our analysis for countries that are on fixed or quasi-fixed exchange regimes. Second, we consider a number of countries that adopted inflation targeting relatively recently. We study the behaviour of *RERs*, *NERs*, and relative inflation rates before and after countries adopt inflation targeting.

3.3.1. Fixed and quasi-fixed exchange rates. In this subsection, we redo our analysis for countries with fixed or quasi-fixed exchange rates. Results for China and Hong Kong, which have quasi-fixed and fixed exchange rates, respectively, are reported in panel (b) of Table 1. Several features of these results are worth noting. First, the estimated values of $\beta_{i,h}^{NER}$ are small relative to the estimates for our benchmark countries. Second, the estimated values of $\beta_{i,h}^{\pi}$ are statistically significant at every horizon and are large relative to the estimates for our benchmark countries. Third, the estimated values of $\beta_{i,h}^{\pi}$ are larger at longer horizons.

We also consider several euro-area countries—France, Ireland, Italy, Portugal, and Spain—vis-à-vis Germany, starting in 1999. For these countries, the *NER* is fixed. Results for regression (3.3) are reported in panel (c) of Table 1. As was the case for China and Hong Kong vis-à-vis the U.S., the estimated values of $\beta_{i,h}^{\pi}$ are large, rise in magnitude with the horizon, and are statistically significant at long horizons.

In sum, for economies with fixed or quasi-fixed exchange rates, the *RER* adjusts overwhelmingly through predictable inflation differentials, not through changes in the *NER*.

3.3.2. Countries with changes in exchange rate policy. In this subsection, we redo our analysis for recent inflation-targeting countries (Brazil, Chile, Colombia, Israel, Mexico, Norway, Peru, Philippines, South Africa, South Korea, and Thailand). We initially consider two sample periods. The first sample goes from 1982:Q4 until the date at which the country adopted inflation targeting (see footnote 7 for these dates). The second sample begins when the country adopted inflation targeting and ends in 2018:Q4. We include the period in which the zero lower bound (ZLB) is binding in the U.S. and in some other countries in order to have enough observations to estimate our regressions at a five-year horizon. Our experience with the benchmark countries suggests that including the ZLB period has a mild effect on the coefficients in regressions (3.2) and (3.3) (see [Supplementary Appendix A](#)).

Tables (2) and (3) report our estimates of $\beta_{i,h}^{NER}$ and $\beta_{i,h}^{\pi}$ using data from the first sample. In contrast to our benchmark results, the estimates of $\beta_{i,h}^{NER}$ and $\beta_{i,h}^{\pi}$ do not follow the consistent pattern observed for the benchmark flexible exchange rate countries. Indeed, there is no apparent pattern across the countries considered.

Tables (2) and (3) also report results for the second sample in which all of the countries have adopted inflation targeting. In sharp contrast to the first-sample results, the estimates of $\beta_{i,h}^{NER}$ are negative and statistically different from zero at longer horizons for every country in our sample. For almost all countries, the absolute value of the coefficients grows with the size of h . By contrast, the estimates of $\beta_{i,h}^{\pi}$ are relatively small in the inflation-targeting sample.

Interestingly, according to Table (2), for various values of h , the estimates of $\beta_{i,h}^{NER}$ based on the first sample-period for Brazil, Israel, Mexico, and Peru are significantly positive. This result turns out to depend on not controlling for changes in monetary policy regimes that occurred before these countries became inflation targeters. Figure 2 displays the inflation rates for all of the recent inflation-targeting countries over the full sample, with the vertical line indicating the date at which a country officially became an inflation targeter. Note that for Brazil, Israel, Mexico, and Peru, there is a very large spike in inflation before the date of the vertical line, with Peru being the most extreme example. The first sample clearly mixes monetary policy regimes. For the

TABLE 2
NER regression results, other countries.

$\beta_{i,h}^{NER}$	Before inflation targeting				During inflation targeting			
	Horizon (in years)				Horizon (in years)			
	1	3	5	7	1	3	5	7
Brazil	1.01 (1.08)	2.33 (3.17)	2.66 (2.53)	6.58 (1.80)	-0.18 (0.12)	-0.64 (0.19)	-1.20 (0.13)	-1.48 (0.07)
Chile	-0.15 (0.23)	-0.77 (0.74)	-1.45 (0.60)	-2.16 (0.23)	-0.34 (0.14)	-1.00 (0.16)	-1.30 (0.13)	-1.35 (0.11)
Colombia	-0.15 (0.10)	-0.72 (0.17)	-1.29 (0.23)	-1.66 (0.19)	-0.16 (0.13)	-0.57 (0.25)	-1.08 (0.19)	-1.38 (0.11)
Israel	2.20 (0.90)	2.88 (1.29)	2.73 (1.01)	2.29 (1.30)	-0.44 (0.13)	-0.67 (0.17)	-1.18 (0.26)	-1.29 (0.29)
Mexico	0.77 (0.37)	0.78 (0.55)	0.60 (0.60)	1.18 (0.77)	-0.23 (0.14)	-0.39 (0.31)	-0.51 (0.46)	-0.29 (0.31)
Norway	-0.17 (0.13)	-0.65 (0.10)	-0.86 (0.13)	-0.94 (0.07)	-0.36 (0.14)	-0.90 (0.21)	-1.68 (0.28)	-1.85 (0.20)
Peru	0.21 (0.41)	2.10 (1.23)	5.48 (1.23)	8.11 (0.95)	-0.26 (0.11)	-0.80 (0.19)	-1.25 (0.14)	-1.64 (0.09)
Philippines	-0.26 (0.22)	-0.68 (0.10)	-1.16 (0.17)	-1.22 (0.13)	-0.11 (0.09)	-0.45 (0.12)	-0.73 (0.10)	-0.91 (0.09)
S. Africa	-0.40 (0.11)	-0.94 (0.23)	-0.96 (0.08)	-1.12 (0.04)	-0.37 (0.10)	-1.29 (0.17)	-1.56 (0.21)	-1.12 (0.13)
S. Korea	-0.32 (0.14)	-1.15 (0.16)	-1.27 (0.12)	-1.27 (0.23)	-0.50 (0.19)	-0.92 (0.27)	-1.20 (0.12)	-0.92 (0.09)
Thailand	-0.49 (0.13)	-2.01 (0.84)	-1.42 (0.82)	-0.81 (0.43)	-0.15 (0.07)	-0.45 (0.13)	-0.74 (0.11)	-0.97 (0.06)

Sources: Authors' calculations; data from International Monetary Fund, International Financial Statistics.

case of Israel, Mexico, and Peru, the fall in inflation after the large spike occurs relatively early in the first sample. So, for these countries we estimate regressions (3.2) and (3.3) for the period between the rough date at which the inflation spike ends and the point at which inflation targeting begins.¹⁰ The results are reported in Table 4. Note that the estimates of $\beta_{i,h}^{NER}$ are very similar to those obtained using the inflation-targeting period. Put differently, as soon as inflation outcomes resemble those obtained under official inflation targeting, the regression results are very similar.

In sum, once we control for the monetary policy regime, the co-movement between *NERs*, inflation and *RERs* is very similar in the recent inflation-targeting countries and the benchmark countries. We view these results as supportive of our hypothesis that the monetary policy regime is a central determinant of the way that the *RER* adjusts to shocks.

3.4. Out-of-sample forecasts

In the previous section, we argued that for our inflation-targeting countries, changes in the *NER* at long horizons display a strong negative correlation with the current level of the *RER*. A potential problem with this result is that if the *RER* is very persistent, we might find in-sample predictability when none is actually present. Here, we focus on the usefulness of the *RER* in out-of-sample forecasting to test the null hypothesis that the *NER* is not predictable. Our key result is that if we control for the monetary policy regime in effect, we can reject this null hypothesis. We show that

10. The relevant sample dates for Israel, Mexico, and Peru are 1988:Q1-1997:Q1, 1990:Q1-2002:Q1, and 1992:Q1-2001:Q4, respectively.

TABLE 3
Relative-price regression results, other countries.

$\beta_{i,h}^{\pi}$	Before inflation targeting Horizon (in years)				During inflation targeting Horizon (in years)			
	1	3	5	7	1	3	5	7
Brazil	-1.21 (1.10)	-2.92 (3.23)	-3.30 (2.53)	-7.23 (1.85)	-0.02 (0.02)	0.01 (0.05)	0.09 (0.03)	0.11 (0.03)
Chile	-0.13 (0.14)	-0.12 (0.47)	0.14 (0.43)	0.69 (0.20)	0.06 (0.02)	0.17 (0.04)	0.17 (0.05)	0.12 (0.04)
Colombia	-0.12 (0.02)	-0.22 (0.05)	-0.12 (0.06)	0.02 (0.12)	-0.03 (0.02)	-0.04 (0.05)	-0.04 (0.03)	-0.06 (0.03)
Israel	-2.37 (0.77)	-3.60 (1.21)	-3.58 (0.96)	-3.18 (1.22)	0.09 (0.04)	0.07 (0.09)	0.01 (0.15)	-0.16 (0.08)
Mexico	-1.11 (0.24)	-1.62 (0.39)	-1.75 (0.53)	-2.12 (0.65)	-0.02 (0.02)	0.01 (0.08)	-0.02 (0.10)	0.07 (0.04)
Norway	-0.09 (0.02)	-0.34 (0.04)	-0.44 (0.04)	-0.43 (0.04)	0.00 (0.02)	0.10 (0.03)	0.23 (0.03)	0.23 (0.02)
Peru	-0.38 (0.42)	-2.62 (1.32)	-6.34 (1.28)	-9.17 (0.98)	0.08 (0.02)	0.22 (0.04)	0.29 (0.05)	0.34 (0.06)
Philippines	-0.10 (0.09)	-0.12 (0.09)	-0.49 (0.10)	-0.51 (0.32)	-0.01 (0.01)	-0.08 (0.03)	-0.15 (0.03)	-0.22 (0.02)
S. Africa	-0.07 (0.08)	-0.19 (0.09)	-0.28 (0.18)	-0.33 (0.21)	-0.05 (0.03)	0.10 (0.03)	0.29 (0.03)	0.18 (0.02)
S. Korea	0.09 (0.03)	0.22 (0.04)	0.18 (0.05)	0.11 (0.08)	-0.00 (0.02)	0.02 (0.05)	0.08 (0.06)	0.07 (0.02)
Thailand	0.04 (0.05)	0.09 (0.26)	-0.21 (0.26)	-0.52 (0.15)	0.00 (0.02)	-0.03 (0.04)	-0.08 (0.04)	-0.10 (0.05)

Sources: Authors' calculations; data from International Monetary Fund, International Financial Statistics.

using information about the *RER* systematically helps forecast the *NER* out of sample, at medium and long horizons.

Our forecasting equation for the *NER* is

$$\log\left(\frac{NER_{i,t+h}}{NER_{i,t}}\right) = \beta_h^{NER} (\log(RER_{i,t}) - \mu_{RER,i}) + \varepsilon_{i,t,t+h}^{NER} \quad (3.4)$$

Note that the parameter β_h^{NER} is common across countries. This specification corresponds to an unbalanced panel with a common slope coefficient.¹¹ The value $\mu_{RER,i}$ is the mean of $\log(RER_{i,t})$, calculated using data from the date a country adopts inflation targeting until date t . Our forecasting equation is exact in a model where symmetric countries follow the same monetary policy in the sense that they would have a common value of β_h^{NER} and the expected steady-state change in the nominal exchange rate is zero. Because we need an initial estimate for $\mu_{RER,i}$, we require that a country be in the sample for at least three years before we include it in the regression analysis. We do not start out-of-sample forecasting until 1993, at which point Germany and New Zealand are in the sample.¹²

We assess our ability to forecast the *NER* relative to a forecast of no change. The latter is the benchmark in the literature and corresponds to the assumption that the *NER* is a random walk

11. In adopting this approach, we follow Mark and Sul (2001), Groen (2005), Engel *et al.* (2007), and Mark and Sul (2011), who use panel methods to improve the forecasting power of exchange-rate models.

12. In practice, quarterly consumer price indexes are available with a one-period lag. To address this potential source of look-ahead bias, we redid all of our analysis with a measure of the *RER* for country i using lagged price indexes. We found that our results are very robust to this change.

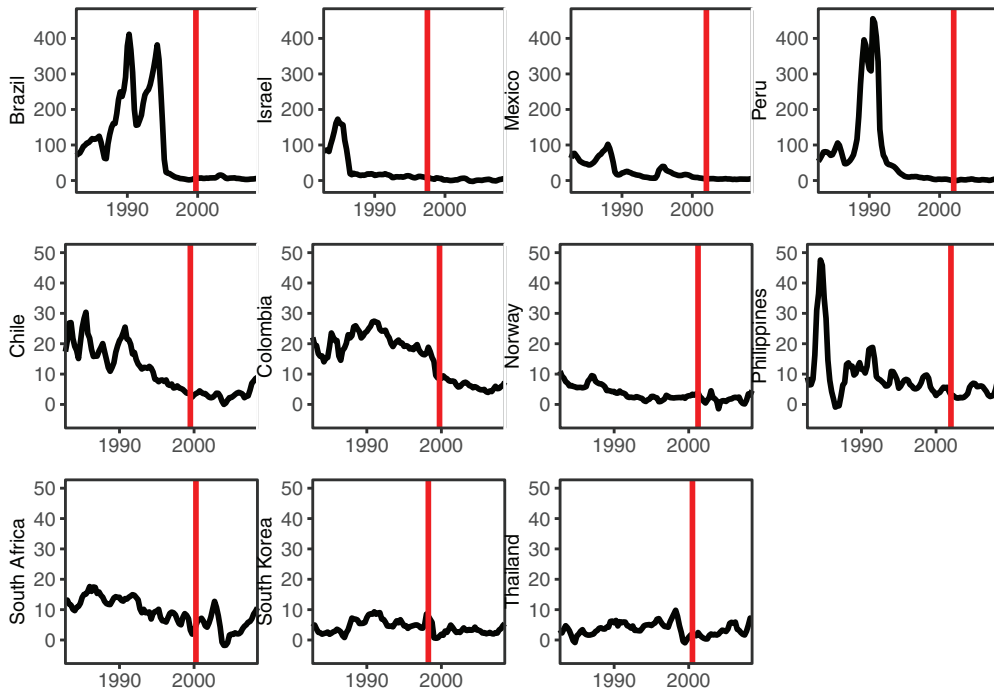


FIGURE 2

Inflation in countries that are recent inflation targeters. The horizontal axis is time. Vertical axis is CPI inflation, computed as the four-quarter log difference. The vertical lines indicate the date at which each country adopted inflation targeting. *Sources:* Data from International Monetary Fund, International Financial Statistics.

TABLE 4
Within-regime regression results

$\hat{\rho}_{i,h}^{NER}$	Horizon (in years)			$\hat{\rho}_{i,h}^{\pi}$	Horizon (in years)		
	1	2	3		1	2	3
Israel	-0.39 (0.29)	-0.88 (0.40)	-1.26 (0.33)	Israel	-0.22 (0.08)	-0.44 (0.17)	-0.18 (0.20)
Mexico	-0.18 (0.21)	-0.98 (0.35)	-1.62 (0.41)	Mexico	-0.40 (0.07)	-0.23 (0.15)	-0.16 (0.18)
Peru	-0.73 (0.40)	-1.26 (0.50)	-1.39 (0.44)	Peru	0.44 (0.29)	0.63 (0.49)	-0.08 (0.62)

Sources: Authors' calculations; data from International Monetary Fund, International Financial Statistics.

without drift. Define the root mean-squared prediction error (RMSPE) for country i associated with forecasts based on equation (3.4) as

$$\sigma_{i,B,h} = \left\{ \frac{1}{T_{i,h}} \sum_{t=0}^{T_{i,h}} \left[f_{i,t,t+h} - \log \left(\frac{NER_{i,t+h}}{NER_{i,t}} \right) \right]^2 \right\}^{1/2}. \quad (3.5)$$

Here, $T_{i,h}$ denotes the number of forecasts for $\log(NER_{i,t+h}/NER_{i,t})$ in our sample, and $f_{i,t,t+h}$ is the forecast of $\log(NER_{i,t+h}/NER_{i,t})$ based on equation (3.4). We denote by $\sigma_{i,RW,h}$ the corresponding RMSPE associated with the no-change forecast from a random-walk model.

For each country i , we report the ratio of the RMSPE associated with the benchmark and random-walk specifications, $\sigma_{i,B,h}/\sigma_{i,RW,h}$. We also compute a pooled RMSPE implied by our forecasting equation for all of the countries in our sample, defined as

$$\sigma_{B,h} = \left\{ \frac{1}{\sum_i T_{i,h}} \sum_i \sum_{t=0}^{T_{i,h}} \left[f_{i,t,t+h} - \log \left(\frac{NER_{i,t+h}}{NER_{i,t}} \right) \right]^2 \right\}^{1/2}. \quad (3.6)$$

We denote by $\sigma_{RW,h}$ the pooled RMSPE implied by the random-walk forecast and report the ratio of the pooled RMSPEs, $\sigma_{B,h}/\sigma_{RW,h}$.

We initially limit the analysis to our benchmark countries. Panel (a) of Table (5) reports relative RMSPEs for each country and for the pooled sample. For the pooled results, forecasts based on equation (3.4) outperform the random-walk model at all horizons greater than two years. Remarkably, at the four- and six-year horizons, forecasting equation (3.4) outperforms the random walk by 24% and 50%, respectively.¹³

We now test the hypothesis that the relative RMSPEs reported in panel (a) of Table 5 were generated by a data generating process (DGP) that implies that the NER is a random walk. Under this hypothesis, changes in the NER should not be predictable. We test this hypothesis using a bootstrap procedure.¹⁴ We assume that the stochastic processes for $NER_{i,t}$ and $RER_{i,t}$ are given by

$$\log \left(\frac{NER_{i,t}}{NER_{i,t-1}} \right) = \varepsilon_{i,t}^{NER}, \quad (3.7)$$

$$A_i(L)(\log(RER_{i,t}) - \mu_{RER,i}) = \varepsilon_{i,t}^{RER}. \quad (3.8)$$

Here, $A_i(L)$ is a polynomial in the lag operator with roots outside the unit circle so that the RER is a stationary process. The parameter $\mu_{RER,i}$ is the mean of $\log(RER_{i,t})$. The random variables $\varepsilon_{i,t}^{NER}$ and $\varepsilon_{i,t}^{RER}$ are uncorrelated over time (though potentially correlated within a period). This DGP embeds the assumption that changes in the NER are unpredictable at all horizons.¹⁵ We consider up to eight lags in $A_i(L)$ and choose the lag length separately for each country using the Akaike information criterion (AIC). Given the estimates of $A_i(L)$, we compute a time series for $\varepsilon_{i,t}^{RER}$ and $\varepsilon_{i,t}^{NER}$ from the data.

We jointly sample the disturbances to preserve contemporaneous correlations between the NER and RER and across countries. We construct 10,000 synthetic time series, each of length equal to the size of our sample, by randomly selecting a sequence of estimated disturbances from the period for which we have a balanced panel. Limiting the bootstrapping to this period preserves the covariance among the shocks.¹⁶ For each synthetic time series, we compute forecasts based on equation (3.4) and the random walk without drift. Using these forecasts, we compute RMSPEs for each country and pooled across countries.

13. Additional recent evidence against random-walk-based forecasts for the NER comes from Cheung *et al.* (2019). Using a sample that spans different monetary policy regimes, they find that for some countries and some sub-samples, relative-PPP-based forecasts outperform the random-walk model.

14. In Supplementary Appendix A, we provide t-statistics from a test studied in Diebold and Mariano (1995) and West (1996). The implications of the t-statistics are broadly similar to our bootstrap p-values. See Rossi (2005) for a discussion of the properties of the Diebold and Mariano (1995) test in an environment similar to ours.

15. Note that if $\log(NER_{i,t}/NER_{i,t-1})$ has a non-zero mean, that property is reflected in the fitted shocks from which we construct the bootstrap samples.

16. We have a burn-in period of 1,000 quarters so that the initial values of $\log(RER_{i,t})$ are different across bootstrap samples.

TABLE 5
Out-of-sample forecasting for the NER.

	Forecast horizon (in years)					
	1	2	3	4	5	6
(a) RMSPE relative to random walk						
All countries	1.08	1.03	0.93	0.76	0.62	0.50
Australia	1.11	1.11	1.06	0.89	0.68	0.45
Canada	1.24	1.34	1.26	1.06	0.89	0.85
Germany	1.04	0.97	0.83	0.64	0.50	0.37
New Zealand	1.05	0.98	0.83	0.66	0.53	0.43
Sweden	1.07	0.98	0.83	0.63	0.50	0.31
U.K.	1.03	0.96	0.86	0.77	0.72	0.72
(b) Bootstrap p-values (stationary <i>RER</i>)						
All countries	0.81	0.39	0.04	0.00	0.00	0.00
Australia	0.89	0.76	0.59	0.13	0.02	0.00
Canada	0.93	0.89	0.76	0.42	0.12	0.09
Germany	0.56	0.14	0.03	0.00	0.00	0.00
New Zealand	0.71	0.21	0.02	0.00	0.00	0.00
Sweden	0.81	0.25	0.05	0.01	0.01	0.00
U.K.	0.40	0.14	0.07	0.04	0.03	0.03
(c) Bootstrap p-values (non-stationary <i>RER</i>)						
All countries	0.33	0.12	0.04	0.00	0.00	0.00
Australia	0.60	0.46	0.38	0.21	0.08	0.02
Canada	0.70	0.63	0.49	0.31	0.18	0.15
Germany	0.32	0.18	0.10	0.03	0.01	0.00
New Zealand	0.38	0.21	0.08	0.02	0.01	0.00
Sweden	0.65	0.36	0.17	0.06	0.04	0.01
U.K.	0.36	0.22	0.14	0.10	0.08	0.10

Sources: Authors' calculations using data from International Monetary Fund, International Financial Statistics.

Panel (b) of Table 5 shows the percentage of bootstrap simulations in which the value of the relative RMSPE is less than or equal to the analogue number reported in panel (a) at different horizons. Consider the first row that pertains to the pooled results. For the three-year horizon, we can reject the random-walk hypothesis at a 5% significance level. At the four-, five-, and six-year horizons, we can reject the random-walk hypothesis at a 1% significance level.

Up to this point, we maintained the assumption that the *RER* is stationary. To assess the robustness of our results, we redo the out-of-sample bootstrap exercises assuming that $\log(RER_{i,t})$ is difference stationary. In particular, we assume that

$$B_i(L)(1-L)\log(RER_{i,t}) = \mu_{\Delta RER,i} + \varepsilon_{i,t}^{RER}. \quad (3.9)$$

Here, $B_i(L)$ is a polynomial in the lag operator with roots outside the unit circle, and $\mu_{\Delta RER,i}$ captures the mean of the log difference of the *RER* for county *i*. We maintain the assumption that changes in the *NER* are given by equation (3.7). As before, we choose the lag length by the AIC and compute the relative RMSPEs. The implied p-values are reported in panel (c) of Table 5. The results are very similar to those reported in panel (b) of that table. We conclude that our results are *not* sensitive to whether we assume that the *RER* has a unit root.

The results in this section provide strong support for the view that the *NER* is forecastable at medium and long horizons. Based on these results, we infer that the in-sample regressions coefficients in the previous section are not spurious.

4. INTERPRETING OUR RESULTS IN A SIMPLE ECONOMIC MODEL

In this section, we use variants of a simple model to highlight the role of inflation targeting in generating our empirical findings. We work with flexible prices to emphasize that the qualitative results from this section do not depend on the presence of nominal rigidities. In the next section, we consider an estimated medium-scale DSGE model that does allow for nominal rigidities. It turns out that the key intuition from our simple model carries over to the estimated model.

4.1. *An stochastic-endowment economy*

In this subsection, we consider an economy with a stochastic endowment. For analytical tractability, the model features a number of simplifying assumptions such as complete markets, UIP, and the law of one price.

4.1.1. Model setup. The model consists of two countries, which we refer to as home and foreign. We refer to the home currency as dollars. The home country is of size n and the foreign country is of size $1 - n$. We express all variables as per-capita values. To conserve space, we display the relevant equations only for the home country. Similar equations, detailed in [Supplementary Appendix B](#), hold for the foreign country. The home country is populated by a representative household with lifetime utility, U , given by

$$U = E_0 \sum_{t=0}^{\infty} \beta^t \log(C_t). \quad (4.10)$$

Here, C_t denotes consumption of the home country, E_t the expectations operator conditional on time- t information, and $0 < \beta < 1$.

Households can trade in a complete set of domestic and international contingent claims. The domestic household's flow budget constraint is given by

$$P_t C_t + B_{\$t} + NER_t B_t \leq P_{Y,t} Y_t + R_{t-1} B_{\$t-1} + NER_t R_{t-1}^* B_{t-1} + T_t. \quad (4.11)$$

Here, P_t is the price of domestic consumption, $B_{\$t}$ and B_t are nominal balances of dollar-denominated and foreign-currency-denominated bonds, R_t and R_t^* are the nominal interest rate on the home and foreign bond, and NER_t is the nominal exchange rate, defined as in our empirical section as the price of the foreign currency unit (units of home currency per unit of foreign currency). The variable T_t denotes nominal lump-sum taxes and net proceeds from contingent claims.¹⁷ With complete markets, the presence of one-period nominal bonds is redundant, as these bonds can be synthesized using state-contingent claims.

The output endowment of the home country, Y_t , follows an exogenous law of motion:

$$\log(Y_t) = \rho_Y \log(Y_{t-1}) + \varepsilon_{Y,t}, \quad (4.12)$$

where $\varepsilon_{Y,t}$ is an i.i.d. shock that follows a normal distribution.

17. Money holdings can easily be added to the model by including a separable additive term in the utility function and modifying the budget constraint accordingly.

The first-order conditions with respect to domestic and foreign bond holdings are

$$1 = \beta R_t E_t \frac{C_t}{\pi_{t+1} C_{t+1}}, \quad (4.13)$$

$$1 = \beta R_t^* E_t \frac{C_t}{\pi_{t+1} C_{t+1}} \frac{NER_{t+1}}{NER_t}, \quad (4.14)$$

respectively. Here, $\pi_t \equiv P_t/P_{t-1}$ denotes the inflation rate in the home country.

The home consumption good, C_t , is produced by combining domestic and foreign intermediate goods ($Y_{H,t}$ and $Y_{F,t}$, respectively) according to the technology

$$C_t = \left[\omega^{1-\rho} Y_{H,t}^\rho + (1-\omega)^{1-\rho} Y_{F,t}^\rho \right]^{\frac{1}{\rho}}. \quad (4.15)$$

Here, $n \leq \omega < 1$ controls the degree of home bias in consumption.¹⁸ The parameter $\rho < 1$ controls the elasticity of substitution between home and foreign goods. Intermediate consumption goods are traded, but final consumption goods are not.

We define the *RER* in units of the home consumption good per one unit of the foreign consumption good,

$$RER_t = \frac{NER_t P_t^*}{P_t}. \quad (4.16)$$

Here, P_t^* is the foreign currency price of the foreign consumption good. With this definition, an increase in RER_t corresponds to a rise in the relative price of the foreign consumption good.

Complete markets and symmetry of initial conditions imply

$$RER_t = \frac{C_t}{C_t^*}. \quad (4.17)$$

Because prices are flexible, the law of one price holds. Market clearing in the intermediate-goods market for the two countries requires $Y_{H,t} + \frac{1-n}{n} Y_{H,t}^* = Y_t$ and $\frac{n}{1-n} Y_{F,t} + Y_{F,t}^* = Y_t^*$. Here, $Y_{H,t}^*$ and $Y_{F,t}^*$ denote the home and foreign output, respectively, used in producing the foreign consumption good. The variable Y_t^* denotes the exogenous endowment of output in the foreign country. This endowment follows an AR(1) process with first-order serial correlation ρ_Y that is analogous to the process described by equation (4.12). Finally, we assume that both home and domestic bonds are in zero net supply.

In the home country, the monetary authority sets the nominal interest rate according to

$$R_t = \beta^{-1} \pi_t^{\theta_\pi}. \quad (4.18)$$

We assume that the Taylor principle holds, so that $\theta_\pi > 1$. We abstract from an output gap term in the rule because prices are flexible. Monetary policy in the foreign country is set in a symmetric manner.

18. The foreign consumption good is produced according to the technology $C_t^* = \left[(\omega^*)^{1-\rho} (Y_{H,t}^*)^\rho + (1-\omega^*)^{1-\rho} (Y_{F,t}^*)^\rho \right]^{\frac{1}{\rho}}$, where $Y_{H,t}^*$ and $Y_{F,t}^*$ are foreign purchases of the home and foreign good, respectively. We assume that the countries are symmetric in the degree of home bias in the sense that $(1-\omega)/(1-n) = (1-\omega^*)/n$.

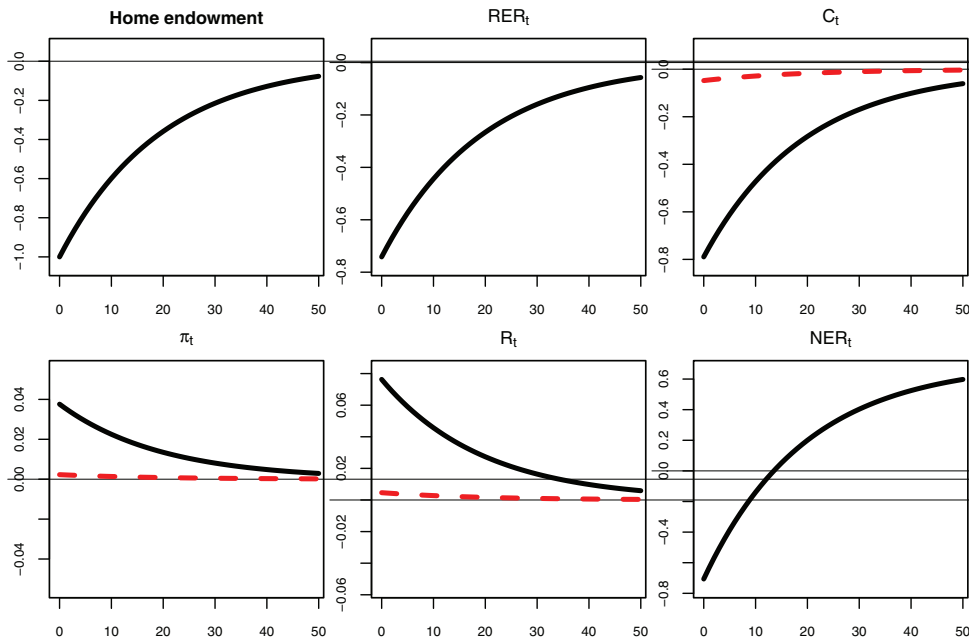


FIGURE 3

Response to endowment shock. The vertical axis is expressed in percent. The horizontal axis shows quarters after the shock. Dashed lines indicate the variables for the foreign country. *Source:* Authors' calculations.

4.1.2. Model properties and regression coefficients. In this section, we use the following parameter values. We set the value of β so that the steady-state real interest rate is 3%. As in Backus *et al.* (1992), we assume that the elasticity of substitution between domestic and foreign goods in the consumption aggregator is 1.5 ($\rho = 1/3$). Motivated by our calibration of the DSGE model in section 5, we assume that the home economy (the U.S.) is 18.5% of the total economy and the import share of the home economy is 12.5% ($\omega = 0.875$). We set ρ_Y , the first-order serial correlation of the endowment, equal to 0.95 and set $\theta_\pi = 2$.

Figure 3 displays the impulse response for a negative shock to the home endowment, Y_t . The RER falls in response to the shock—i.e., the foreign consumption basket becomes cheaper relative to the domestic consumption basket. Home bias plays a critical role in the RER movement. Recall that the RER is given by equation (4.17). So, the RER falls, reflecting the scarcity of home goods and the fact that the home consumption basket places a larger weight on home goods than the size of the home country ($\omega > 0.185$). Put differently, home consumption falls by more than foreign consumption because households in the home country consume more of the good that has become relatively scarce. If there was no home bias ($\omega = 0.185$), the RER would not change in response to the negative shock to Y_t .

Given the differential paths of consumption in the home and foreign country, household Euler equations imply that the domestic real interest rate must be higher than the foreign real interest rate. The Taylor rule and the Taylor principle imply that high real interest rates are associated with high nominal interest rates and high inflation rates. It follows that the nominal interest rate and the inflation rate in the home country rise by more than in the foreign country. This result is inconsistent with the naive intuition that inflation has to be lower in the home country for the RER to return to its pre-shock level. In fact, inflation is persistently *higher* in the home country. So, the RER reverts back to its steady-state value via changes in the NER , not via differential

properties thereafter. As a result, the plims of the regression coefficients, which relate future changes in the *NER* or relative prices to the current level of the *RER*, are independent of ω .

The sum of the two plims is given by

$$\beta_h^{NER} + \beta_h^\pi = -(1 - \rho_Y^h). \quad (4.22)$$

This sum converges to -1 as $h \rightarrow \infty$, reflecting the fact that the *RER* must eventually converge to its pre-shock steady-state level. While this sum converges to -1 , β_h^{NER} converges to a value that is lower than -1 . These properties reflect the fact that the *NER* must eventually adjust by more than the *RER* to bring the latter back to its steady-state value.

In [Supplementary Appendix B](#), we analyse a version of our simple model in which the Taylor rule in the foreign country is modified so that policy makers place some weight on stabilizing the *NER*. The policy rule takes the form

$$R_t^* = \beta^{-1} (\pi_t^*)^{\theta_\pi} NER_t^{-\theta_{NER}}$$

We show that our qualitative analysis is robust to allowing for small values of θ_{NER} .

4.2. Shocks to the foreign demand for dollar-denominated bonds

In this subsection, we analyse the effect of a shock to the foreign demand for dollar-denominated assets. The intuition for the impact of this shock is most easily provided in a setting with complete domestic markets and in which the only internationally traded asset is a dollar-denominated bond. The latter assumption is consistent with the evidence in [Maggiore et al. \(2020\)](#). We also use this asset market specification in our estimated DSGE model (see Section 5). All notation is as above, except for new variables.

4.2.1. Model setup. The home country is populated by a representative household with lifetime utility, U , given by (4.10). The expected lifetime utility, U^* , of the representative household in the foreign country is given by

$$U^* = E_0 \sum_{t=0}^{\infty} \beta^t \left[\log(C_t^*) + \log(\tilde{\eta}) V \left(\frac{B_{\$t}^*}{P_t^*} NER_t^{-1} \right) \right]. \quad (4.23)$$

Here, $B_{\$t}^*$ are the foreign household's holdings of dollar-denominated bonds.

The function V governs the utility flow from dollar-denominated bond holdings. The variable $\tilde{\eta}_t$ is a shock to the utility that the foreign household derives from holding those bonds. For convenience, we assume that $\log(\tilde{\eta}_t)$ is zero in steady state, so the steady-state utility flow from the bonds is zero. Outside of steady state, there may be shocks, reflecting flights to safety or liquidity concerns, that put a premium on dollar-denominated bonds. This type of shock breaks UIP in log-linear versions of the model. Instead of introducing a shock directly into the UIP condition, as in [McCallum \(1994\)](#), we assume that households derive utility from bond holdings and that this utility flow varies over time. [Itkhoki and Mukhin \(2017\)](#) provide an extensive discussion of the micro foundations of these shocks, which involves the activities of noise traders. Another motivation for these shocks comes from [Fisher \(2015\)](#), who emphasizes the safety and liquidity properties of bonds.

The budget constraints of the home and foreign households are given by

$$P_t C_t + B_{\$t} \leq R_{t-1} B_{\$t-1} + P_{Y,t} Y_t \quad (4.24)$$

$$P_t^* C_t^* + B_{\$t}^* NER_t^{-1} + \Phi_{B,t} + B_t^* \leq NER_t^{-1} R_{t-1} B_{\$t-1}^* + R_{t-1}^* B_{t-1}^* + P_{Y,t}^* Y_t^*. \quad (4.25)$$

As above, consumption is created by combining intermediate goods from the home and foreign country according to equation (4.15). The term $\Phi_{B,t}$ denotes the costs to holding dollar-denominated bonds. These adjustment costs induce stationarity of the *RER* (see [Schmitt-Grohé and Uribe, 2003](#)). We assume that these costs are very small.²⁰ The home and foreign goods follow the same endowment process as in Section 4.1.

In [Supplementary Appendix B](#), we formally define a competitive equilibrium for this model economy. We also show that the effect of an endowment shock in this model is very similar to the effect of an endowment shock in a version of this model with complete markets.

4.2.2. Model properties and regression coefficients. We compute impulse response functions and regression coefficients using the same parameter values discussed in Section 4.1. We assume that $\log(\tilde{\eta}_t)$ follows an AR(1) process with an autoregressive coefficient equal to 0.95.

Figure 4 displays the impulse responses to an increase in the foreign demand for dollar-denominated bonds, $\tilde{\eta}_t$. This shock leads to an increase in the demand for dollar-denominated bonds and the dollars required to pay for those bonds. So, the dollar appreciates (*i.e.* the *NER* falls) and the interest rate on dollar-denominated bonds (R_t) falls. The appreciation of the dollar implies that foreign goods are cheaper for the home household, so P_t falls. The Taylor rule implies that the fall in inflation is relatively small, so the *RER* also falls (the home *RER* appreciates). Because R_t falls and the home demand for dollar-denominated bonds is not directly affected by $\tilde{\eta}_t$, the home country finances a rise in consumption by borrowing from the foreign country. Equivalently, the home household supplies more of the bonds that the rest of the world is demanding. To buy those bonds, foreigners reduce their consumption.

In the period after the shock, $\tilde{\eta}_t$ begins to return to its pre-shock level and the dollar depreciates over time. So, a low level of the *RER* in the period of the shock is associated with a future increase in the *NER*. This pattern generates negative values for β_h^{NER} . As was the case for the endowment shock in Section 4.1, β_h^{NER} is increasing in absolute value with h because the cumulative appreciation of the foreign currency increases over time. As shown in Figure 5, in the endowment economy considered in this sub-section with shocks only to $\tilde{\eta}_t$, the plims of β_h^{NER} are negative and less than one in absolute value. Recall that, in the endowment economy of the previous sub-section that included only endowment shocks, for large h , the plims of β_h^{NER} all exceed one. So, we anticipate that the magnitudes of the plims of β_h^{NER} in an estimated version of the model depend on the parameter values.

Two other properties of our simple model are worth noting. First, UIP does not hold in this simple model. Figure 4 implies that in response to an increase in $\tilde{\eta}_t$, R_t is less than R_t^* . However, the dollar depreciates over time, the opposite of the pattern implied by UIP. If shocks to $\tilde{\eta}_t$ are quantitatively important, standard tests would reject UIP. Second, in response to an $\tilde{\eta}_t$ shock, our simple model implies that C_t/C_t^* co-moves negatively with the *RER*. Recall that in the endowment economy, the endowment shock leads C_t/C_t^* to co-move positively with the *RER*.

20. We assume that $\Phi_{B,t} = \Phi_B \left(\frac{B_{\$t}^* NER_t^{-1}}{P_t^*} \right) P_t^*$, where $\Phi_B(x) = \frac{0.001}{2} x^2$.

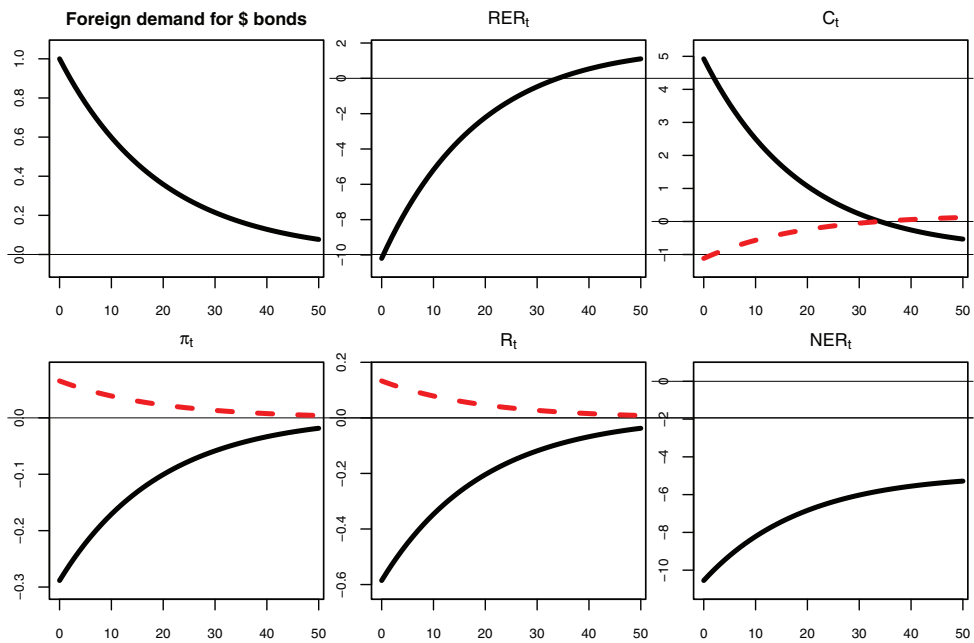


FIGURE 4

Response to shock to foreign demand for dollar-denominated bonds. The vertical axis is expressed in percent. The horizontal axis shows quarters after the shock. Dashed lines indicate the variables for the foreign country. *Source:* Authors' calculations.

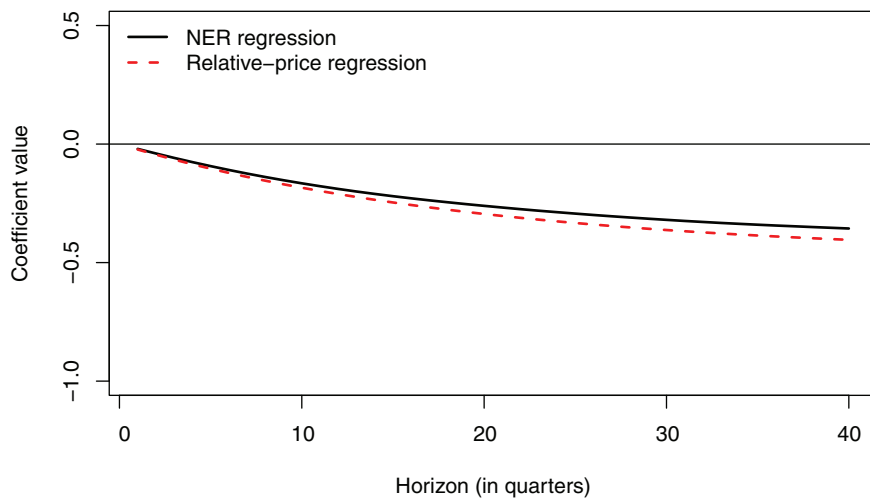


FIGURE 5

β_h^{NER} and β_h^{π} with shocks to foreign demand for dollar-denominated bonds.

So, if shocks to $\tilde{\eta}_t$ are quantitatively important, C_t/C_t^* may not be strongly positively correlated with the *RER*—*i.e.*, the model may not exhibit the [Backus and Smith \(1993\)](#) puzzle.

5. ESTIMATED MEDIUM-SCALE DSGE MODEL

In this section, we address the following question: which shocks and frictions account quantitatively for the observed correlation between the current *RER* and future changes in the *NER*? We answer this question by considering an estimated medium-scale DSGE model with three regions: the U.S., Germany, and the rest of the world. The model incorporates the two key features of our simple endowment economy: home bias and inflation targeting in the form of a Taylor rule. Our analysis focuses on fluctuations in the bilateral exchange rate between the U.S. and Germany. Having three regions allows us to study bilateral exchange rates without having to make implausible assumptions about import and export shares.

5.1. Households

For notational ease, the U.S., Germany, and the rest of the world correspond to country 1, 2, and 3, respectively. Each country i has a continuum of households of size $n_i \in (0, 1)$. The size of the world population is equal to one: $n_1 + n_2 + n_3 = 1$. As in [Christiano *et al.* \(2005\)](#), each household makes three sets of decisions per period. First, each household decides how much to consume, how much capital to accumulate, and how much capital services to supply to the market. Second, each household purchases securities, whose payoffs are contingent on whether it can re-optimize its nominal wage rate. We assume that there are complete contingent claims markets within each country. Only U.S. dollar-denominated bonds can be traded internationally. Third, each household sets its nominal wage rate after finding out whether it can re-optimize it.

Because households face idiosyncratic risk about whether they can re-optimize their nominal wage rates, hours worked and wage rates differ across households. So, in principle, households are heterogeneous with respect to consumption and asset holdings. It follows from a straightforward extension of arguments in [Woodford \(1998\)](#) and [Erceg *et al.* \(2000\)](#) that, in equilibrium, households in a given country are homogeneous with respect to consumption and asset holdings. Reflecting this result, our notation assumes that households are homogeneous with respect to consumption and asset holdings but heterogeneous with respect to their wage rate and hours worked.

Similar to [Christiano *et al.* \(2005\)](#), we assume that the utility of household k in country i is given by

$$U_{k,i} = E_0 \sum_{t=0}^{\infty} \beta^t \mu_{i,t} \left[\log(C_{i,t} - f \bar{C}_{i,t-1}) - \frac{\chi}{2} L_{i,t}(k)^2 + \sum_{j=1}^3 \log(\eta_{i,j,t}) V\left(\frac{B_{i,j,t}}{P_{i,t}} NER_{i,j,t}\right) \right]. \quad (5.26)$$

Here, $C_{i,t}$ is the consumption of each household in country i and $\bar{C}_{i,t}$ is per-capita aggregate consumption. $L_{i,t}(k)$ are hours worked by household k in country i . The scalar f controls the degree of habit formation in preferences. $B_{i,j,t}$ is the end-of-period- t holdings of country- j bonds held by the households in country i . $P_{i,t}$ is the consumer price index in country i denominated in local currency units. $NER_{i,j,t}$ is the price of country- j currency in units of country- i 's currency. The variable $\mu_{i,t}$ is a shock to the household's discount rate.

As in the endowment economy, the function V governs the utility flow from bond holdings of different countries. The variable $\eta_{i,j,t}$ is a shock to the utility that country i derives from holding the bonds of country j . For convenience, we assume that $\eta_{i,j,t}$ is one in steady state, so that the steady-state utility flow from bonds is zero.²¹ Our assumptions about $\eta_{i,j,t}$, detailed below, allow for the possibility of world-wide shocks to the marginal utility of holding dollar-denominated bonds or shocks to that marginal utility that affect only non-U.S. households.

In our model, there are three types of bonds corresponding to the three currencies, with only the dollar-denominated bond traded internationally. The household budget constraint of country i is

$$\sum_{j=\{1,i\}} B_{i,j,t} \text{NER}_{i,j,t} + P_{i,t} C_{i,t} + P_{i,t} I_{i,t} + a_i(u_{i,t}) \bar{K}_{i,t} P_{i,t} + \varphi_{B,i,t} = \sum_{j=\{1,i\}} R_{j,t-1} B_{i,j,t-1} \text{NER}_{i,j,t} + R_{i,t}^K u_{i,t} \bar{K}_{i,t} + W_{i,t}(h) L_{i,t}(h) + T_{i,t}, \quad (5.27)$$

where

$$\varphi_{B,i,t} = \mathbf{1}\{i \neq 1\} \Phi_{1,B} \left(\frac{B_{i,1,t} \text{NER}_{i,1,t}}{P_{i,t}} \right) P_{i,t}, \quad (5.28)$$

$R_{i,t}^K$ is the rental rate on capital in country i , $\bar{K}_{i,t}$ is the stock of capital owned by the households in country i , $I_{i,t}$ is investment in country i , $u_{i,t}$ is the capital utilization rate, $u_{i,t} \bar{K}_{i,t}$ denotes the period- t supply of capital services, and $a_i(u_{i,t}) \bar{K}_{i,t}$ denotes the cost of capital utilization. $T_{i,t}$ are net receipts from all contingent claims of the household as well as lump-sum taxes, transfers, and profits received from domestic firms. The function $\Phi_{i,B}$ represents the costs of holding foreign bonds. As in [Schmitt-Grohé and Uribe \(2003\)](#), we include these costs to avoid the presence of a unit root in the equilibrium process for the RER . We assume that there are no costs of holding bonds denominated in the home currency. The functional forms for a_i and $\Phi_{1,B}$ are described below.

We model nominal wage rigidities as in [Erceg et al. \(2000\)](#). A labour aggregator combines labour services from each household to produce the homogeneous labour input used in production, $L_{i,t}$, according to

$$L_{i,t} = \left(\frac{1}{n_i} \int_0^{n_i} (L_{i,t}(h))^{\frac{v_{i,t}-1}{v_{i,t}}} dh \right)^{\frac{v_{i,t}}{v_{i,t}-1}}. \quad (5.29)$$

Household h is a monopoly supplier of $L_{i,t}(h)$. The variable $W_{i,t}(h)$ represents the wages paid to household h . Labour aggregators are perfectly competitive and take the nominal wage for the homogeneous labour input, $W_{i,t}$, as given. With probability $1 - \xi_{W,i}$, a household updates its wage rate to maximize its utility. With probability $\xi_{W,i}$, the wage grows at its steady-state growth rate. The random variable $v_{i,t}$ controls the substitution between labor types.

The capital accumulation equation is

$$\bar{K}_{i,t+1} = \zeta_{i,t} F_i(I_{i,t}, I_{i,t-1}) + (1 - \delta) \bar{K}_{i,t}, \quad (5.30)$$

where the variable $\zeta_{i,t}$ is an investment-specific technology shock. The function F_i embeds the technology that transforms current and past investment into capital. We discuss the properties of F below. The parameter δ controls the capital depreciation rate.

21. We could accommodate steady-state differences in interest rates net of expected exchange-rate movements between countries of the type stressed by [Hassan and Mano \(2018\)](#) by allowing the unconditional mean of $\log(\eta_{i,j,t})$ to be different from zero.

The household creates the final good by combining intermediate goods from each of the three countries according to the production function

$$Y_{i,t} = \left(\sum_{j=1}^3 \omega_{i,j}^{1-\rho} [\varphi_{i,j,t} Y_{i,j,t}]^\rho \right)^{\frac{1}{\rho}}. \quad (5.31)$$

Here, $Y_{i,j,t}$ denotes purchases of wholesale goods produced from country j . The price of $Y_{i,j,t}$ in country i 's currency is $P_{i,j,t}$. The parameters $\omega_{i,j}$ control the importance of intermediate goods from country j in producing $Y_{i,t}$, and $\sum \omega_{i,j} = 1$. The term $\varphi_{i,j,t}$ represents adjustment costs associated with changing the ratio of imports to domestically produced goods. As in [Erceg *et al.* \(2006\)](#), we assume that

$$\varphi_{i,j,t} = \left[1 - \frac{\varphi_i}{2} \left(\frac{Y_{i,j,t}/Y_{i,i,t}}{Y_{i,j,t-1}/Y_{i,i,t-1}} - 1 \right)^2 \right]. \quad (5.32)$$

[Erceg *et al.* \(2006\)](#) argue that these adjustment costs enable the model to capture the relatively sluggish response to shocks to the share of imports in final goods.

5.2. Producers

The wholesale good, $Y_{i,j,t}$, is produced by perfectly competitive wholesalers using a continuum of intermediate goods according to the technology

$$Y_{i,j,t} = \left(\frac{1}{n_j} \right)^{\frac{1}{v_{j,t}}} \left(\int_0^{n_j} X_{i,j,t}(m)^{\frac{v_{j,t}-1}{v_{j,t}}} dm \right)^{\frac{v_{j,t}}{v_{j,t}-1}}. \quad (5.33)$$

Here, $X_{i,j,t}(m)$ denotes purchases of the m th intermediate good from country j by the wholesaler in country i . The random variable $v_{j,t}$ controls the substitution between intermediate goods. As this variable is indexed by j , we are assuming that a country- j producer of intermediate good m is affected by $v_{j,t}$ regardless of where the good is sold.

The intermediate good $X_{i,j,t}(m)$ is produced by a monopolist in country j using the technology

$$A_{j,t} K_{j,t}^\alpha (L_{j,t}(m))^{1-\alpha} = \sum_{i=1}^3 X_{i,j,t}(m). \quad (5.34)$$

The variables $K_{j,t}(m)$ and $L_{j,t}(m)$ denote the amount of capital and labor hired by monopolist m in country j . The intermediate good producers set their price in the currency where their goods are sold (so-called “local-currency pricing”). With probability $1 - \xi_{P,i}$, monopolist m sets prices, $P_{j,i,t}(m)$, for $j = 1, 2, 3$ to maximize profits, which are given by

$$E_0 \sum_{t=0}^{\infty} \beta^t \Lambda_{i,t} \left(\frac{NER_{i,j,t} P_{j,i,t}(m)}{P_{i,t}} - MC_{i,t} \right) X_{j,i,t}(m), \quad (5.35)$$

subject to demand for the product. The variable $\Lambda_{i,t}$ is the marginal utility of the household in country i during period t , and $MC_{i,t}$ is the monopolist's real marginal cost of producing $X_{i,j,t}(m)$. With probability $\xi_{P,i}$, monopolists increase their prices by the steady-state inflation rate in country j .

5.3. Monetary policy

In country i , the monetary authority follows a Taylor rule given by

$$\frac{R_{i,t}}{R_i} = \left(\frac{R_{i,t-1}}{R_i} \right)^{\gamma_i} \left(\left(\frac{\pi_{i,t}}{\pi_i^*} \right)^{\theta_{\pi,i}} \left(\frac{GDP_{i,t}}{\tilde{GDP}_{i,t}} \right)^{\theta_{GDP,i}} \right)^{1-\gamma_i} \exp(\varepsilon_{R,i,t}) \quad (5.36)$$

where $\theta_{i,\pi} > 1$. Here, $\varepsilon_{R,i,t}$ is a monetary policy shock, R_i is the steady-state nominal interest rate in country i , and π_i^* is the target rate of inflation. $GDP_{i,t}$ is defined as the sum of consumption, investment, government purchases, net exports, and adjustment costs. $\tilde{GDP}_{i,t}$ is the natural level of $GDP_{i,t}$, defined as the level of $GDP_{i,t}$ that would prevail under flexible prices with inflation at its target level. Throughout, we assume that the Taylor principal is satisfied so that $\theta_{\pi,i} > 1$. We also assume that $0 \leq \gamma_i < 1$.

5.4. Clearing of final-good and bond markets

Market clearing for final good $Y_{i,t}$ implies

$$C_{i,t} + G_{i,t} + I_{i,t} + a_i(u_{i,t})\bar{K}_{i,t} + \mathbf{1}\{i \neq 1\} \Phi_{B,1} \left(\frac{B_{i,1,t} NER_{i,1,t}}{P_{i,t}} \right) = Y_{i,t}, \quad (5.37)$$

where $G_{i,t}$ are government purchases of goods in country i . We assume that the government balances its budget in each period with lump-sum taxes. As a result, dollar-denominated bonds are in zero net supply, so

$$\sum_{j=1}^3 n_j B_{j,1,t} = 0. \quad (5.38)$$

We adopt a standard sequence-of-markets equilibrium concept. We work with a standard log-linear approximation around the symmetric balanced-growth steady state. In [Supplementary Appendix C](#), we derive and display the equations that characterize the equilibrium for our model economy.

5.5. Stochastic processes

In this section, we describe our assumptions about the shocks impacting the environment. Even though we allow for many shocks, it turns out that only a small subset of the shocks are quantitatively important drivers of the equilibrium exchange rate. Our approach allows us to identify these shocks.

In what follows, $\varepsilon_{\cdot,i,t}$ are i.i.d. normal random variables and $|\rho_{\cdot,i}| < 1$. The aggregate technology shock $A_{i,t}$ in equation (5.34) follows a trend-stationary process with a global stochastic component (A_t) and a country-specific component ($\tilde{A}_{i,t}$). In particular, we assume that

$$A_{i,t} = \tilde{A}_{i,t} A_t \Upsilon^{t(1-\alpha)}. \quad (5.39)$$

The variable Υ is the unconditional growth rate in a balanced-growth equilibrium.

We assume that A_t ; $\tilde{A}_{i,t}$; the shock to the discount rate for each household, $\mu_{i,t}$, in equation (5.26); the investment-specific technology shock, $\zeta_{i,t}$, in equation (5.30); and the government

purchases shock, $G_{i,t}$, in equation (5.37) evolve according to

$$\log(x_{i,t}) = \rho_{x,i} \log(x_{i,t-1}) + \varepsilon_{x,i,t}. \quad (5.40)$$

Here, $x_{i,t}$ is the ratio of the variable to its steady-state value. The variables $\frac{v_{i,t}}{v_{i,t-1}}$ and $\frac{v_{j,t}}{v_{j,t-1}}$, which act as wage and price markup shocks, also evolve according to (5.40).

The shocks, $\eta_{i,j,t}$, evolve according to

$$\eta_{i,j,t} = \tilde{\eta}_{i,j,t} \eta_{j,t}. \quad (5.41)$$

Recall that $\eta_{i,j,t}$ is a shock to the marginal utility in country i from holding a country- j bond. According to our specification, a change in $\eta_{i,j,t}$ can reflect a change in the marginal utility of a country- i household for country- j bonds (through $\tilde{\eta}_{i,j,t}$) or a world-wide increase in the marginal utility of holding country- j bonds (through $\eta_{j,t}$). The random variables $\eta_{j,t}$ and $\tilde{\eta}_{i,j,t}$ evolve according to equation (5.40). We assume that only $\tilde{\eta}_{2,1,t}$ and $\tilde{\eta}_{3,1,t}$ are potentially different from one. For simplicity, we assume that $\tilde{\eta}_{2,1,t} = \tilde{\eta}_{3,1,t} = \tilde{\eta}_t$. For all other combinations of i and j , $\tilde{\eta}_{i,j,t}$ is one. These assumptions allow us to identify $\eta_{j,t}$ and reflect the special role that U.S. bonds play in the model.

Because we only include global output as an observable variable for the rest of the world (see below), we set a number of shocks in the third country to zero. In particular, we set $\mu_{3,t}$, $\xi_{3,t}$, $G_{3,t}$, $v_{3,t}$, $v_{3,t}$, and $\eta_{3,t}$ to their unconditional steady-state values. So, the only shocks originating from the rest of the world are $\tilde{\eta}_{3,1,t}$ and $\tilde{A}_{3,t}$ —*i.e.*, shocks to the demand for dollar-denominated bonds and technology.

5.6. Functional forms

As in [Christiano *et al.* \(2005\)](#), we assume the function form for investment adjustment costs is given by

$$F_i(I_{i,t}, I_{i,t-1}) = I_{i,t} \left(1 - S_i \left(\frac{I_{i,t}}{I_{i,t-1}} \Upsilon \right) \right), \quad (5.42)$$

where $S_i(1) = S'_i(1) = 0$ and $S''_i(1) > 0$. These properties of S_i are the only ones relevant for the log-linear equilibrium conditions. We assume that $a_i(1) = 0$ and that $u_{i,t} = 1$ in steady state. The only other feature of a that is relevant in the log-linear equilibrium conditions is $a''_i(1)/a'_i(1) > 0$, which we treat as a parameter to be estimated.

We assume that the function V is increasing and strictly concave. The only property of V that is relevant for the log-linear equilibrium conditions is $V'(0)$, which we set equal to the steady-state value of $\Lambda_{i,t}$. This assumption amounts to a normalization according to which $\log(\eta_{i,j,t})$ enters the intertemporal Euler equation with a coefficient of unity.

Finally, we assume that the cost of holding bonds, $\Phi_{B,1}$, is given by

$$\Phi_{B,1}(b) = \frac{\psi_b \Upsilon^t}{2} \left(\frac{b}{\Upsilon^t} \right)^2. \quad (5.43)$$

In our economy, shocks cause borrowing and lending among countries. The magnitude of that borrowing and lending reflects growth in economy-wide variables, like real GDP. We scale real bond holdings by Υ^t so that the adjustment costs do not rise as the economy grows.

6. ESTIMATION

In this section, we accomplish four tasks. First, we discuss the data used in our analysis and our estimation procedure. Second, we discuss parameters that we fix *a priori* and the parameters that we estimate using Bayesian methods as in [An and Schorfheide \(2007\)](#). Third, we analyse which shocks and frictions account *quantitatively* for the movements in the *RER* and the *NER* as well as their covariance with inflation. Fourth, we consider how the economy would have behaved under alternative monetary policy regimes.

6.1. Data

We estimate the model using U.S. and German data for the following variables: the demeaned growth rate of per-capita consumption, GDP, and investment; the real wage; the short-term interest rate; the rate of inflation; and the U.S.–German exchange rate. We also include data on hours worked in the U.S., employment for Germany, and global GDP. See [Supplementary Appendix D](#) for a detailed description of our data series.

6.2. Model parameters

We set the markup parameters to $v_i = 21$ and $v_i = 6$, which are in the range considered by [Altig et al. \(2011\)](#). We calibrate the steady-state ratio of government purchases to output to 0.18. Consistent with the literature, we set $\alpha = 0.25$, $f = 0.75$, $\beta = 0.9968$, and $\delta = 0.025$.

We set Υ , the unconditional quarterly growth rate of output, to the average quarterly growth rate of per-capita output across the U.S. and Germany in our sample (1.0046). As we are working with a log-linear version of the model, the target inflation rates for the U.S., Germany, and the rest of the world do not affect the empirical properties of the model.

Because of data limitations, we set the following rest-of-the-world parameters to the common mean of the priors for the corresponding U.S. and German parameters: $S_3''(1) = 4$, $\frac{a_3''(1)/a_3'(1)}{a_3'(1)/a_3'(1)+1} = 0.5$, $\gamma_3 = 0.75$, $\theta_{\pi,3} = 1.7$, $\theta_{GDP,3} = 0.1$, and $\varphi_3 = 10$. In addition, we set the Calvo parameters ($\xi_{P,3}$ and $\xi_{W,3}$) so that prices and wages are optimized, on average, once per year.

The remaining parameters are estimated using standard Bayesian methods. Tables 6, 7, and 8 display our prior distributions, as well as the posterior mode, posterior standard deviation, and the interval between the 5th and 95th percentile of the posterior distribution.²² Several features are worth noting. First, the model estimates imply fairly standard values for the Taylor rule coefficient for both the U.S. and Germany. Second, the posterior distributions of shock variances are less dispersed than the prior distributions. Third, there are some differences between the parameter estimates specific to the U.S. and the parameter estimates specific to Germany. In particular, the persistence of $\zeta_{2,t}$ (the marginal efficiency of investment) is much lower in Germany than in the U.S., while the persistence of technology shocks is higher in Germany.

6.3. Analysis

We next turn to the question, which shocks and frictions account *quantitatively* for the movements in the *RER* and the *NER* as well as their covariance with inflation?²³ We begin by quantifying

22. We use a Metropolis–Hastings algorithm to simulate draws from the posterior distribution of the parameters. We draw two chains of length 1,000,000 from the posterior distribution and discard the first 500,000 draws from each chain. We then combine the two chains.

23. Throughout this section, statistics are computed for the logarithm of the relevant variables.

TABLE 6
Posterior distribution: non-shock parameters

		Prior Shape (mean, std. dev.)	Mode	Posterior st. dev.	Posterior [5%, 95%]
U.S.					
γ_1	Taylor rule persistence	Beta (0.75, 0.05)	0.87	0.01	[0.85, 0.89]
$\theta_{\pi,1} - 1$	Taylor rule infl. coef.	Inv. Gamma (0.7, 0.15)	0.68	0.14	[0.52, 0.96]
$\theta_{GDP,1}$	Taylor rule GDP coef.	Beta (0.10, 0.05)	0.04	0.02	[0.02, 0.08]
$\xi_{P,1}$	Calvo price setting	Beta (0.5, 0.10)	0.33	0.05	[0.24, 0.41]
$\xi_{W,1}$	Calvo wage setting	Beta (0.5, 0.10)	0.55	0.06	[0.50, 0.70]
$S_1''(1)$	Investment adj. costs	Normal (4.00, 1.50)	5.24	1.10	[3.80, 7.41]
$\frac{a_1''(1)/a_1'(1)}{a_1''(1)/a_1'(1)+1}$	Utilization adj. costs	Beta (0.50, 0.15)	0.85	0.06	[0.73, 0.93]
φ_1	Net export adj. costs	Normal (10, 2)	12.72	1.80	[9.75, 15.68]
Germany					
γ_2	Taylor rule persistence	Beta (0.75, 0.05)	0.92	0.01	[0.90, 0.93]
$\theta_{\pi,2} - 1$	Taylor rule infl. coef.	Inv. Gamma (0.7, 0.15)	0.49	0.07	[0.41, 0.64]
$\theta_{GDP,2}$	Taylor rule GDP coef.	Beta (0.10, 0.05)	0.26	0.05	[0.18, 0.35]
$\xi_{P,2}$	Calvo price setting	Beta (0.5, 0.10)	0.76	0.05	[0.67, 0.83]
$\xi_{W,2}$	Calvo wage setting	Beta (0.5, 0.10)	0.72	0.06	[0.60, 0.80]
$S_2''(1)$	Investment adj. costs	Normal (4.00, 1.50)	3.88	0.95	[2.66, 5.78]
$\frac{a_2''(1)/a_2'(1)}{a_2''(1)/a_2'(1)+1}$	Utilization adj. costs	Beta (0.50, 0.15)	0.63	0.12	[0.40, 0.81]
φ_2	Net export adj. costs	Normal (10, 2)	9.53	2.03	[6.19, 12.89]

Notes: In our Bayesian estimation, we simulate two MCMC chains of length 1,000,000. We discard the first 500,000 draws from each chain, and we combine the remaining observations from the two chains. Source: Authors' calculations.

which shocks drive the volatility of the *RER* between the U.S. and Germany. Table 9 reports the fraction of the variance of the *RER* accounted for by the most important shocks. The bulk (75%) of the variation in the *RER* is accounted for by the shock to foreign demand for dollar-denominated bonds ($\tilde{\eta}_t$). So, from the perspective of the model, the majority of the variance in the *RER* arises from $\tilde{\eta}_t$ shocks that affect the difference in yields between U.S. and foreign bonds. No other single shock accounts for more than 5% of the variance of the *RER*.

Table 10 reports data- and model-based statistics regarding U.S.–German exchange rates. Three features of Table 10 are worth noting. First, our model does a reasonable job of matching the volatilities of both ΔRER and ΔNER and is consistent with the fact that real and nominal exchange rates are equally volatile. Second, the model is consistent with the classic observation of Mussa (1986) that changes in *RERs* and *NERs* are highly correlated. Third, the model implies that the *RER* is highly persistent, though somewhat less so than in the data.

We now show that our estimated model can quantitatively account for the estimated values of $\beta_{i,h}^{NER}$ and $\beta_{i,h}^{\pi}$ implied by regressions (3.2) and (3.3). To this end, we simulate our estimated model, drawing shocks from the estimated distributions, and run those regressions on the simulated data. The row labelled “Asymptotic value” in Table 11 refers to regressions run on a time series of length 1,000,000 and the row labelled “Small sample” refers to the mean of the estimates from 10,000 regressions run on simulated time series, each of length 100. The row labelled “Small-sample st. dev.” refers to the standard deviation of the estimates across the 10,000 regressions. Note that the model accounts for the negative values of $\beta_{i,h}^{NER}$, which grow in absolute value with the horizon. The estimated values of $\beta_{i,h}^{\pi}$ are all small in absolute value and statistically insignificantly different from zero.

A key question is, which shock, in practice, accounts for the covariance between the *RER* and future changes in the *NER*? This covariance is related to the regression coefficient $\beta_{i,h}^{NER}$ in

TABLE 7
Posterior distribution: shock process persistence.

		Mode	St. dev.	[5%, 95%]
U.S.				
$\rho_{A,1}$	Technology	0.93	0.03	[0.86, 0.97]
$\rho_{\zeta,1}$	Marginal efficiency of investment	0.82	0.07	[0.68, 0.90]
$\rho_{v,1}$	Price markup	0.94	0.02	[0.90, 0.97]
$\theta_{v,1}$	Price markup	0.24	0.09	[0.09, 0.38]
$\rho_{v,1}$	Wage markup	0.81	0.11	[0.50, 0.86]
$\theta_{v,1}$	Wage markup	0.97	0.03	[0.89, 0.98]
$\rho_{\mu,1}$	Marginal utility of consumption	0.92	0.03	[0.86, 0.95]
$\rho_{G,1}$	Government purchases	0.96	0.01	[0.93, 0.98]
Germany				
$\rho_{A,2}$	Technology	0.98	0.02	[0.94, 0.99]
$\rho_{\zeta,2}$	Marginal efficiency of investment	0.05	0.05	[0.02, 0.17]
$\rho_{v,2}$	Price markup	0.87	0.07	[0.68, 0.92]
$\theta_{v,2}$	Price markup	0.70	0.15	[0.27, 0.78]
$\rho_{v,2}$	Wage markup	0.94	0.03	[0.86, 0.96]
$\theta_{v,2}$	Wage markup	0.96	0.02	[0.91, 0.97]
$\rho_{\mu,2}$	Marginal utility of consumption	0.09	0.07	[0.03, 0.25]
$\rho_{G,2}$	Government purchases	0.91	0.04	[0.83, 0.96]
Rest of world				
$\rho_{A,3}$	Technology	0.97	0.02	[0.92, 0.99]
Foreign				
$\rho_{\bar{\eta}}$	Demand for dollar bonds	0.87	0.02	[0.83, 0.90]
Global				
ρ_A	Technology	0.56	0.20	[0.21, 0.88]
$\rho_{\eta,1}$	Demand for dollar bonds	0.22	0.13	[0.08, 0.50]
$\rho_{\eta,2}$	Demand for German bonds	0.79	0.05	[0.71, 0.86]

Notes: We use a Beta prior with mean 0.5 and standard deviation 0.2 for all parameters shown in this table. Source: Authors' calculations.

equation (3.2) via the relationship

$$\beta_{i,h}^{NER} = \frac{\text{cov}(RER_{i,t}, \Delta^h NER_{i,t+h})}{\text{var}(RER_{i,t})} = \frac{\sum_{\epsilon} \text{cov}(RER_{i,t}^{\epsilon}, \Delta^h NER_{i,t+h}^{\epsilon})}{\text{var}(RER_{i,t})}, \quad (6.44)$$

where the sum is over the shocks. Table 12 reports the results of decomposing this covariance by shock. At all horizons reported, over 70% of the negative covariance is due to the shock to foreign demand for dollar-denominated bonds.

We now turn to the performance of the model on other dimensions of the data. Tables 13 and 14 report the model-implied decomposition of the variation in U.S. and German GDP. The key result is that shocks to the foreign demand for dollar-denominated bonds account for small percentages of the variance in GDP. For the U.S., the key shocks are similar to those highlighted by Smets and Wouters (2007) and Justiniano *et al.* (2011)—*e.g.*, shocks to the marginal efficiency of investment as well as technology and markup shocks. According to the model, technology shocks drive the bulk of the variance in German real GDP. Consistent with the closed economy literature, shocks to monetary policy only play a small role in accounting for variation in GDP (see *e.g.* Smets and Wouters, 2007). This result does not imply that the systematic component of monetary policy is unimportant in the behaviour of GDP. In fact, a key point of this paper is

TABLE 8
Posterior distribution: shock process volatility

		Mode	st. dev.	[5%, 95%]
U.S.				
$\sigma_{A,1}$	Technology	0.55	0.04	[0.49, 0.63]
$\sigma_{\zeta,1}$	Marginal efficiency of investment	0.42	0.06	[0.37, 0.56]
$\sigma_{R,1}$	Monetary policy	0.16	0.01	[0.14, 0.19]
$\sigma_{v,1}$	Price markup	1.42	0.43	[1.01, 2.39]
$\sigma_{\nu,1}$	Wage markup	0.40	0.03	[0.36, 0.47]
$\sigma_{\mu,1}$	Marginal utility of consumption	0.30	0.04	[0.25, 0.37]
$\sigma_{G,1}$	Government purchases	0.42	0.03	[0.37, 0.48]
Germany				
$\sigma_{A,2}$	Technology	0.67	0.07	[0.57, 0.79]
$\sigma_{\zeta,2}$	Marginal efficiency of investment	1.16	0.09	[1.02, 1.32]
$\sigma_{R,2}$	Monetary policy	0.10	0.01	[0.09, 0.11]
$\sigma_{v,2}$	Price markup	0.47	0.06	[0.37, 0.56]
$\sigma_{\nu,2}$	Wage markup	0.69	0.06	[0.63, 0.82]
$\sigma_{\mu,2}$	Marginal utility of consumption	0.46	0.04	[0.41, 0.55]
$\sigma_{G,2}$	Government purchases	0.64	0.06	[0.56, 0.76]
Rest of world				
$\sigma_{A,3}$	Technology	0.51	0.11	[0.43, 0.77]
Foreign				
$\sigma_{\tilde{\eta}}$	Demand for dollar bonds	0.10	0.02	[0.08, 0.15]
Global				
σ_A	Technology	0.08	0.03	[0.06, 0.16]
$\sigma_{\eta,1}$	Demand for dollar bonds	0.06	0.01	[0.04, 0.09]
$\sigma_{\eta,2}$	Demand for German bonds	0.07	0.02	[0.05, 0.11]

Notes: We use an Inverse Gamma prior with mean 0.2 and standard deviation 1 for all parameters shown in this table.

Source: Authors' calculations.

TABLE 9
Variance decomposition of US/German RER

		Percent
$\tilde{\eta}_t$	Foreign demand for dollar bonds	75
$\mu_{1,t}$	U.S. marginal utility of consumption	5
$\eta_{2,t}$	German demand for German bonds	4
$A_{2,t}$	German technology	4
$\zeta_{1,t}$	U.S. marginal efficiency of investment	3
$A_{1,t}$	U.S. technology	2
$v_{1,t}$	U.S. price markup	2
$\nu_{1,t}$	U.S. wage markup	2

Notes: Statistics are computed using parameter values from the posterior mode. Shocks are excluded from the table if they account for less than 2% of the variance at each frequency. *Source:* Authors' calculations.

that the effect of shocks on exchange rates and the rest of the economy depends critically on the monetary policy regime—*i.e.* the systematic component of monetary policy.

Taken together, our results imply that shocks to foreign demand for dollar-denominated bonds are an important driver of the *RER* without being an important driver of output fluctuations. This finding is consistent with the arguments in [Itskhoki and Mukhin \(2017\)](#). To provide further evidence on the relation between exchange rates and fluctuations in real variables, Table 15 reports the correlation between ΔNER with the growth rates of consumption, real GDP, and investment in the U.S. and Germany. We also report the correlation between trade-weighted ΔNER and

TABLE 10
Exchange rate statistics

	Data	Model
St. dev. ΔRER	4.63% (0.27)	5.31%
St. dev. ΔNER	4.69% (0.27)	5.44%
Correlation $\Delta NER, \Delta RER$	0.99 (0.00)	0.98
Autocorrelation US/GE RER	0.96 (0.03)	0.85

Notes: Statistics are computed using parameter values from the posterior mode of our benchmark model. Standard errors for data estimates are GMM standard errors. *Source:* Authors' calculations.

TABLE 11
Regression coefficients in estimated DSGE model

	Horizon (in years)			
	1	3	5	7
(a) NER regression coefficient, $\beta_{i,h}^{NER}$				
Asymptotic value	−0.46	−0.86	−1.00	−1.06
Small-sample mean	−0.56	−0.97	−1.09	−1.11
Small-sample st. dev.	0.17	0.28	0.35	0.42
(b) Relative-price regression coefficient, $\beta_{i,h}^{\pi}$				
Asymptotic value	0.01	0.04	0.06	0.07
Small-sample mean	0.00	0.01	0.02	0.01
Small-sample st. dev.	0.07	0.19	0.28	0.36

Notes: Statistics are computed using parameter values from the posterior mode. “Asymptotic value” refers to regressions run on a time series of length 1,000,000. “Small-sample mean” refers to the mean of the estimates from 10,000 regressions run on simulated time series, each of length 100. “Small-sample st. dev.” refers to the standard deviation of the estimates across the 10,000 regressions. *Source:* Authors' calculations.

TABLE 12
Shocks driving NER regression coefficients in estimated DSGE model

		Horizon (in years)			
		1	3	5	7
$\tilde{\eta}_t$	Foreign demand for dollar bonds	75	75	74	72
$\mu_{1,t}$	U.S. marginal utility of consumption	3	5	6	6
$\eta_{2,t}$	German demand for German bonds	8	6	5	5
$\zeta_{1,t}$	U.S. marginal efficiency of investment	2	3	4	4
$A_{2,t}$	German technology	2	3	3	4
$v_{1,t}$	U.S. wage markup	2	2	2	3
$v_{1,t}$	U.S. price markup	2	2	2	2
$A_{1,t}$	U.S. technology	2	2	2	2

Notes: This table shows the percent of the correlation between the RER and future changes in the NER accounted for by each shock at different horizons. Statistics are computed using parameter values from the posterior mode. *Source:* Authors' calculations.

U.S. net exports.²⁴ The growth rates of U.S. consumption, investment, and GDP all display a mild negative correlation with ΔNER . In contrast, those correlations are all slightly positive for

24. We do not have data on U.S. net exports to Germany. So, to calculate the relevant correlations, we used total U.S. net exports. We calculated the correlation of the change in real net exports and the log change of the FRED series TWEXBMTH for the trade-weighted U.S. exchange rate. In the model, we calculate the trade-weighted U.S. exchange

TABLE 13
Variance decomposition of U.S. GDP

		Percent
$\zeta_{1,t}$	U.S. marginal efficiency of investment	41
$v_{1,t}$	U.S. wage markup	16
$A_{1,t}$	U.S. technology	15
$v_{1,t}$	U.S. price markup	15
$G_{1,t}$	U.S. government purchases	4
$\mu_{1,t}$	U.S. marginal utility of consumption	3
$R_{1,t}$	U.S. monetary policy	3
$\tilde{\eta}_t$	Foreign demand for dollar bonds	2

Notes: Statistics are computed using parameter values from the posterior mode. Shocks are excluded from the table if they account for less than 2% of the variance at each frequency. Source: Authors' calculations.

TABLE 14
Variance decomposition of German GDP

		Percent
$A_{2,t}$	German technology	68
$\eta_{2,t}$	German demand for German bonds	11
$v_{2,t}$	German wage markup	6
$G_{2,t}$	German government purchases	4
$R_{2,t}$	German monetary policy	2
$v_{2,t}$	German price markup	2
$A_{3,t}$	Rest-of-world technology	2
$\zeta_{2,t}$	German marginal efficiency of investment	2
$\mu_{2,t}$	German marginal utility of consumption	2

Notes: Statistics are computed using parameter values from the posterior mode. Shocks are excluded from the table if they account for less than 2% of the variance at each frequency. Source: Authors' calculations.

TABLE 15
Correlations with the NER in the data and in the estimated model

Correlation	$\Delta C_{1,t}$	$\Delta I_{1,t}$	$\Delta GDP_{1,t}$	$\Delta NX_{1,t}$	$\Delta C_{2,t}$	$\Delta I_{2,t}$	$\Delta GDP_{2,t}$
Model	−0.08	−0.02	0.26	0.17	−0.08	−0.05	−0.22
Data	−0.23 (0.08)	−0.18 (0.07)	−0.15 (0.08)	0.20 (0.12)	0.09 (0.07)	−0.04 (0.09)	0.01 (0.10)

Notes: Model statistics are computed using parameter values from the posterior mode of our benchmark model. Standard errors (in parentheses) are Newey–West standard errors with 4 lags. All correlations are computed with $\Delta NER_{1,2,t}$ except for $\Delta NX_{1,t}$, where we use the trade-weighted exchange rate. Source: Authors' calculations.

Germany. With the exception of U.S. GDP, our model does a good job of accounting for both the U.S. and German correlations, taking sampling uncertainty into account. Our model produces a counterfactually strong correlation (0.26) between the growth rate in U.S. GDP and ΔNER . Finally, the model accounts well for the mild positive correlation between the change in U.S. net exports and ΔNER .

In discussing the simple flexible-price-endowment economy of Section 4, we modelled changes in the RER as arising from home bias. With sticky prices, the RER can respond to shocks even if there is no home bias. We explore the role of home bias per se by setting the weight on each good in equation (5.31) equal to the corresponding size of each country. So, for

rate by weighting the $\Delta NER_{1,2,t}$ and $\Delta NER_{1,3,t}$ by U.S. steady-state trade shares. We could not get reliable German net export data for the entire sample that distinguishes between German net exports to the euro area and to the rest of the world.

TABLE 16
Regression coefficients in DSGE models with alternative specifications

	Horizon (in years)			
	1	3	5	7
(a) <i>NER</i> regression coefficient, $\beta_{i,h}^{NER}$				
No home bias	−0.63	−0.90	−0.97	−0.99
<i>NER</i> targeting	−0.28	−0.38	−0.39	−0.37
Capital controls	−0.48	−0.88	−1.02	−1.06
(b) Relative-price regression coefficient, $\beta_{i,h}^{\pi}$				
No home bias	−0.21	−0.11	−0.04	−0.01
<i>NER</i> targeting	−0.02	−0.24	−0.42	−0.52
Capital controls	0.01	0.05	0.06	0.07

Notes: Statistics are computed using parameter values from the posterior mode of our benchmark model. Regressions are run on a time series of length 1,000,000. Source: Authors' calculations.

example, $\omega_{11} = 0.185$, $\omega_{12} = 0.054$, and $\omega_{13} = 0.761$. With this specification there is no home bias. The model-implied values of β_h^{NER} and β_h^{π} are shown in Table 16. The key result is that the absolute values of the model-implied plims of β_h^{NER} are slightly smaller in absolute value than in the estimated model. This result makes clear that sticky prices have a role to play in explaining the dynamic correlations between *NER*s and *NER*s. But home bias, or alternatively non-traded goods, has a separate role to play.

We now provide two additional pieces of evidence in favour of the model's empirical credibility. The first pertains to the “Backus–Smith puzzle.” Backus and Smith (1993) document that the *RER* is at best weakly correlated with relative consumption across countries. However, many models counterfactually imply a high, positive correlation between the *RER* and relative consumption. As explained in Section 4, other things equal, shocks to $\tilde{\eta}_t$ induce a negative correlation between these variables. To see whether our model is subject to the Backus–Smith puzzle, we simulate it using the fitted disturbances of all the shocks and run the following regression:

$$\widehat{RER}_{1,2,t} = a_0 + a_1 (\hat{C}_{1,t} - \hat{C}_{2,t}) + \epsilon_t. \quad (6.45)$$

The plim of a_1 implied by our model is equal to 0.08. The estimated value of a_1 in our data sample is 0.07 with a standard error of 0.25. So, taking sampling uncertainty into account, our estimated model is quantitatively consistent with the observed co-movement between the *RER* and relative consumption.

We now turn to the model's implications for the “forward-premium puzzle” originally documented by Fama (1984). Many models counterfactually imply that the interest rate differential predicts changes in the *NER*. This property is easily seen from a log-linear version of equations (4.13) and (4.14), which imply that the expected change in a bilateral *NER* is equal to the interest rate differential on bonds denominated in the two currencies—*i.e.*, UIP holds. According to our estimated model, UIP does not hold conditional on shocks to $\tilde{\eta}_t$, $\eta_{1,t}$, or $\eta_{2,t}$, but it holds for many of the other shocks in our model, like disturbances to technology. To determine whether our model can quantitatively account for the failure of the standard, unconditional version of UIP, we simulated the estimated model using the fitted shocks. We then run the “Fama regression” on the simulated time series:

$$\Delta \widehat{NER}_{1,2,t+1} = d_0 + d_1 (\hat{R}_{1,t} - \hat{R}_{2,t}) + \epsilon_t. \quad (6.46)$$

The plim of d_1 implied by our model is equal to 0.17. The estimated value of d_1 in our data sample is -0.67 with a standard error of 0.74. So, taking sampling uncertainty into account, our model is quantitatively consistent with the observed co-movements between interest rate differentials and exchange rates.

6.4. Exploring alternative monetary policy regimes

In this subsection, we consider how the economy would have behaved under two alternative monetary policy regimes. In the first regime, the monetary authority gives some weight to stabilizing the *NER*. We refer to this policy as the *NER*-targeting regime. In the second regime, there are capital controls, which we model as large costs of holding foreign bonds (a large value of ψ_b in equation (5.43)).

In the *NER*-targeting regime, the German monetary policy rule is given by

$$\frac{R_{2,t}}{R_2} = \left(\frac{R_{2,t-1}}{R_2} \right)^{\gamma_2} \left(\left(\frac{\pi_{2,t}}{\pi_2^*} \right)^{\theta_{\pi,2}} \left(\frac{GDP_{2,t}}{\tilde{GDP}_{2,t}} \right)^{\theta_{GDP,2}} \right)^{1-\gamma_2} NER_{2,1,t}^{\theta_{NER,2}} \exp(\varepsilon_{R,2,t}), \quad (6.47)$$

where $\theta_{NER,i} > 0$ and $\pi_2^* = \pi_1^*$. According to equation (6.47), the monetary authority raises the nominal interest rate when the domestic currency depreciates. The larger is $\theta_{NER,2}$, the higher the weight the monetary authority gives to stabilizing the exchange rate. As $\theta_{NER,2} \rightarrow \infty$, this rule converges to a fixed exchange rate regime.

To assess the impact of the change in monetary policy on exchange rates, we focus on the implications for $\beta_{i,h}^{NER}$ and $\beta_{i,h}^{\pi}$. To this end, we generate artificial time series from our estimated model but assume that German monetary policy is governed by equation (6.47) with $\theta_{NER,2} = 0.1$. The latter assumption implies that a 10% depreciation of the German currency is accompanied by a 1% quarterly increase in the German nominal interest rate. Table 16 reports the model-implied plims for $\beta_{i,h}^{NER}$ and $\beta_{i,h}^{\pi}$. The values of $\beta_{i,h}^{NER}$ are much smaller than in our benchmark model, and the values of $\beta_{i,h}^{\pi}$ are much larger. These results imply that the *RER* is adjusting, over time, primarily through differential inflation rates, not through changes in the *NER*. The reason is that the monetary policy authority does let the *NER* change by as much as it does in the estimated model.

Under the *NER*-targeting regime, differential inflation rates play a much larger role in reestablishing long-run PPP. Interestingly, the persistence of the *RER*, as measured by its first-order auto correlation, rises from 0.85 under inflation targeting to 0.90 under the *NER*-targeting regime. In this sense, inflation targeting results in more rapid adjustment of the *RER* to its long-run value.

To capture the effects of capital controls, we assume that the cost to German households of holding dollar-denominated bonds (ψ_b) is 100 times larger than in our estimated model. The larger is the cost of holding dollar-denominated bonds, the closer the German economy is to financial autarky. Under our assumption, the peak rise in German holdings of dollar-denominated bonds after a shock to $\tilde{\eta}_t$ is less than 10% of what it is in our benchmark estimated model. The volatility of the ΔRER and ΔNER each falls by about 3% relative to the benchmark model. Shocks to the foreign demand for dollar-denominated bonds account for roughly the same percentage of the unconditional variance of U.S. and German real GDP (about 2% and 0%, respectively). Finally, we find that model-implied $\beta_{i,h}^{NER}$ and $\beta_{i,h}^{\pi}$ are similar to the estimates obtained in our benchmark model. In this sense, the latter estimates are robust to allowing for capital controls.

7. CONCLUSION

This article shows that in inflation-targeting countries, the *RER* adjusts to shocks in the medium and long run through changes in the *NER*, not via differences in inflation rates. For such countries, the current *RER* is useful for forecasting future changes in the *NER*. Consistent with the Lucas (1976) critique, these facts depend critically on the monetary policy regime in effect.

Using a simple endowment economy, we provide intuition for the economics underlying our statistical findings. This intuition relies on two key assumptions: the monetary authority follows an inflation-targeting policy like a Taylor rule and the *RER* moves in response to various shocks. Home bias in consumption is one way to generate movements in the *RER*. Taylor rules are important because they keep inflation relatively stable and cause relative prices to move in such a way so that the *NER* has to adjust by more than the *RER* over time.

We build and estimate a medium-scale, open-economy DSGE model to answer the question, which shocks and frictions account *quantitatively* for the movements in the *RER* and the *NER* as well as their covariance with inflation? We find that shocks to the foreign demand for dollar-denominated bonds drive the bulk of exchange rate movements. These shocks also quantitatively account for the dynamic correlations that drive the predictability of the *NER*. We argue that our inferences are credible because the model reproduces key empirical facts about the *NER* and *RER* as well as aggregate economic fluctuations.

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Supplementary Data

Supplementary data are available at *Review of Economic Studies* online.

REFERENCES

- ALTIG, D., CHRISTIANO, L. J., M. EICHENBAUM et al. (2011), "Firm-Specific Capital, Nominal Rigidities and the Business Cycle", *Review of Economic Dynamics*, **14**, 225–247.
- AMADOR, M., BIANCHI, J., L. BOCOLA et al. (2019), "Exchange Rate Policies at the Zero Lower Bound", *The Review of Economic Studies*, forthcoming.
- AN, S. and SCHORFHEIDE, F. (2007), "Bayesian Analysis of DSGE Models", *Econometric Reviews*, **26**, 113–172.
- BACKUS, D. K., KEHOE, P. J. and KYDLAND, F. E. (1992), "International Real Business Cycles", *Journal of Political Economy*, **100**, 745–775.
- BACKUS, D. K. and SMITH, G. W. (1993), "Consumption and Real Exchange Rates in Dynamic Economies with Non-traded Goods", *Journal of International Economics*, **35**, 297–316.
- BAXTER, M. and STOCKMAN, A. C. (1989), "Business Cycles and the Exchange-Rate Regime: Some International Evidence", *Journal of Monetary Economics*, **23**, 377–400.
- BERNANKE, B. S. and MIHOV, I. (1997), "What Does the Bundesbank Target?", *European Economic Review*, **41**, 1025–1053.
- BOUDOUKH, J., RICHARDSON, M. and WHITELAW, R. F. (2006), "The Myth of Long-Horizon Predictability", *The Review of Financial Studies*, **21**, 1577–1605.
- CHEUNG, Y.-W., CHINN, M. D., PASCUAL, A. G. et al. (2019), "Exchange Rate Prediction Redux: New Models, New Data, New Currencies", *Journal of International Money and Finance*, **95**, 332–362.
- CHEUNG, Y.-W., LAI, K. S. and BERGMAN, M. (2004), "Dissecting the PPP Puzzle: The Unconventional Roles of Nominal Exchange Rate and Price Adjustments", *Journal of International Economics*, **64**, 135–150.
- CHRISTIANO, L. J., EICHENBAUM, M. and EVANS, C. L. (2005), "Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy", *Journal of political Economy*, **113**, 1–45.
- CHRISTIANO, L. J., TRABANDT, M. and VALENTIN, K. (2011), "Introducing Financial Frictions and Unemployment Into a Small Open Economy Model", *Journal of Economic Dynamics and Control*, **35**, 1999–2041.
- CLARIDA, R., GALI, J. and GERTLER, M. (1998), "Monetary Policy Rules in Practice: Some International Evidence", *European Economic Review*, **42**, 1033–1067.

- COCHRANE, J. H. (2007), "The Dog that Did Not Bark: A Defense of Return Predictability", *The Review of Financial Studies*, **21**, 1533–1575.
- DIEBOLD, F. X. and MARIANO, R. S. (1995), "Comparing Predictive Accuracy", *Journal of Business & Economic Statistics*, **13**, 253–263.
- DORNBUSCH, R. (1976), "Expectations and Exchange Rate Dynamics", *Journal of Political Economy*, **84**, 1161–1176.
- ENGEL, C. (2019), "Real Exchange Rate Convergence: The Roles of Price Stickiness and Monetary Policy", *Journal of Monetary Economics*, **103**, 21–32.
- ENGEL, C., MARK, N. C. and WEST, K. D. (2007), "Exchange Rate Models are Not as Bad as You Think", *NBER Macroeconomics Annual*, **22**, 381–473.
- ENGEL, C. and WU, S. P. Y. (2019), "Liquidity and Exchange Rates: An Empirical Investigation" (NBER Working Paper No. 25397).
- ERCEG, C. J., GUERRIERI, L. and GUST, C. (2006), "SIGMA: A New Open Economy Model for Policy Analysis", *International Journal of Central Banking*, **2**, 1–50.
- ERCEG, C. J., HENDERSON, D. W. and LEVIN, A. T. (2000), "Optimal Monetary Policy with Staggered Wage and Price Contracts", *Journal of Monetary Economics*, **46**, 281–313.
- FAMA, E. F. (1984), "Forward and Spot Exchange Rates", *Journal of Monetary Economics*, **14**, 319–338.
- FISHER, J. D. (2015), "On the Structural Interpretation of the Smets–Wouters "Risk Premium" Shock", *Journal of Money, Credit and Banking*, **47**, 511–516.
- GABAIX, X. and MAGGIORI, M. (2015), "International Liquidity and Exchange Rate Dynamics", *The Quarterly Journal of Economics*, **130**, 1369–1420.
- GROEN, J. J. (2005), "Exchange Rate Predictability and Monetary Fundamentals in a Small Multi-country Panel", *Journal of Money, Credit and Banking*, **37**, 495–516.
- HASSAN, T. A. and MANO, R. C. (2018), "Forward and Spot Exchange Rates in a Multi-currency World", *The Quarterly Journal of Economics*, **134**, 397–450.
- HENDERSON, D. W. and MCKIBBIN, W. J. (1993), "A Comparison of Some Basic Monetary Policy Regimes for Open Economies: Implications of Different Degrees of Instrument Adjustment and Wage Persistence", in: *Carnegie-Rochester Conference Series on Public Policy*, Vol. 39 (Amsterdam: North-Holland) 221–317.
- ILZETZKI, E., REINHART, C. M. and ROGOFF, K. S. (2017), "The Country Chronologies to Exchange Rate Arrangements into the 21st Century: Will the Anchor Currency Hold?" (NBER Working Paper No. 23135).
- ILZETZKI, E., REINHART, C. M. and ROGOFF, K. S. (2019), "Exchange Arrangements Entering the 21st Century: Which Anchor Will Hold?", *The Quarterly Journal of Economics*, **134**, 599–646.
- ITSKHOKI, O. and MUKHIN, D. (2017), "Exchange Rate Disconnect in General Equilibrium" (NBER Working Paper No. 23401).
- JIANG, Z., KRISHNAMURTHY, A. and LUSTIG, H. (2019), "Foreign Safe Asset Demand and the Dollar Exchange Rate" (NBER Working Paper, no. 24439).
- JUSTINIANO, A., PRIMICERI, G. E. and TAMBALOTTI, A. (2011), "Investment Shocks and the Relative Price of Investment", *Review of Economic Dynamics*, **14**, 102–121.
- KOLLMANN, R. (2005), "Macroeconomic Effects of Nominal Exchange Rate Regimes: New Insights Into the Role of Price Dynamics", *Journal of International Money and Finance*, **24**, 275–292.
- KOLLMANN, R., PATARACCHIA, B., RACIBORSKI, R. et al. (2016), "The Post-Crisis Slump in the Euro Area and the US: Evidence from an Estimated Three-Region DSGE Model", *European Economic Review*, **88**, 21–41.
- LUBIK, T. and SCHORFHEIDE, F. (2005), "A Bayesian Look at the New Open Economy Macroeconomics", *NBER Macroeconomics Annual*, **20**, 313–366.
- LUCAS, R. E. (1976), "Econometric Policy Evaluation: A Critique", *Carnegie-Rochester Conference Series on Public Policy*, **1**, 19–46.
- MAGGIORI, M., NEIMAN, B. and SCHREGER, J. (2020), "International Currencies and Capital Allocation", *Journal of Political Economy*, **128**, 2019–2066.
- MARK, N. C. (1995), "Exchange Rates and Fundamentals: Evidence on Long-Horizon Predictability", *The American Economic Review*, **85**, 201–218.
- MARK, N. C. and SUL, D. (2001), "Nominal Exchange Rates and Monetary Fundamentals: Evidence from a Small Post-Bretton Woods Panel", *Journal of International Economics*, **53**, 29–52.
- MARK, N. C. and SUL, D. (2011), "When are Pooled Panel-Data Regression Forecasts of Exchange Rates More Accurate than the Time-Series Regression Forecasts?", *Handbook of Exchange Rates*, Chapter 9, 265–281.
- MCCALLUM, B. T. (1994), "A Reconsideration of the Uncovered Interest Parity Relationship", *Journal of Monetary Economics*, **33**, 105–132.
- MEESE, R. A. and ROGOFF, K. (1983), "Empirical Exchange Rate Models of the Seventies: Do They Fit Out of Sample?", *Journal of International Economics*, **14**, 3–24.
- MOLODTSOVA, T. and PAPELL, D. H. (2009), "Out-of-Sample Exchange Rate Predictability with Taylor Rule Fundamentals", *Journal of International Economics*, **77**, 167–180.
- MUSSA, M. (1986), "Nominal Exchange Rate Regimes and the Behavior of Real Exchange Rates: Evidence and Implications", *Carnegie-Rochester Conference Series on Public Policy*, **25**, 117–214.
- NEWKEY, W. K. and WEST, K. D. (1987), "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix", *Econometrica*, **55**, 703–708.
- ROGOFF, K. (1996), "The Purchasing Power Parity Puzzle", *Journal of Economic Literature*, **34**, 647–668.

- ROSSI, B. (2005), "Testing Long-Horizon Predictive Ability with High Persistence, and the Meese-Rogoff Puzzle", *International Economic Review*, **46**, 61–92.
- ROSSI, B. (2013), "Exchange Rate Predictability", *Journal of Economic Literature*, **51**, 1063–1119.
- SARNO, L. and VALENTE, G. (2006), "Deviations from Purchasing Power Parity under Different Exchange Rate Regimes: Do they Revert and, if so, How?", *Journal of Banking & Finance*, **30**, 3147–3169.
- SCHMITT-GROHE, S. and URIBE, M. (2003), "Closing Small Open Economy Models", *Journal of International Economics*, **61**, 163–185.
- SMETS, F. and WOUTERS, R. (2007), "Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach", *American Economic Review*, **97**, 586–606.
- STAMBAUGH, R. F. (1999), "Predictive Regressions", *Journal of Financial Economics*, **54**, 375–421.
- TAYLOR, A. M. and TAYLOR, M. P. (2004), "The Purchasing Power Parity Debate", *Journal of Economic Perspectives*, **18**, 135–158.
- WEST, K. D. (1996), "Asymptotic Inference about Predictive Ability", *Econometrica*, **64**, 1067–1084.
- WOODFORD, M. (1998), "Control of the Public Debt: A Requirement for Price Stability?", in Calvo, G. and King, M. (eds) *The Debt Burden and Its Consequences for Monetary Policy* (UK: Palgrave Macmillan) 117–158.